



Review

Machine learning applications in nanomaterials: Recent advances and future perspectives

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ABSTRACT

Nanomaterials demonstrate enormous potential applications in various scientific and engineering fields due to their unique physical and chemical properties. With the rapid development of machine learning (ML) technology, its role in the application of nanomaterials is becoming increasingly prominent. Nanomaterials, assisted by various ML algorithms, efficiently model their structure–property relationships, enabling precise prediction and rational design. This review aims to explore the state-of-the-art and future trends of ML in nanomaterial research. It focuses on analyzing research strategies for ML-assisted nanomaterials, including design, characterization, and preparation strategies. The review systematically examines research outcomes in property prediction, structure optimization, synthesis design, characterization analysis, image processing, and quality control, while also summarizing and looking ahead to future development directions. The ML not only accelerates the discovery and development of nanomaterials but also enhances the understanding of nanoscale phenomena, broadens the practical applications of nanoscience, and provides new ideas and technological means for intelligent, high-throughput nanomaterial research and development.

Abbreviations: A3C, Asynchronous advantage actor-critic; AFM, Atomic force microscopy; AgNCs, Silver nanoclusters; AI, Artificial intelligence; ANN, Artificial neural network; BPNN, Backpropagation neural network; BR, Bayesian regularization; BSA, Bionic sensor array; BT, Boosted tree; CART, Classification and regression tree; CDs, Carbon dots; CNN, Convolutional neural network; CNT, Carbon nanotube; CV, Crystal violet; DBSCAN, Density-based spatial clustering of applications with noise; DCNN, Deep convolution neural network; DFT, Density functional theory; DNN, Deep neural network; DQN, Deep Q-learning network; DRL, Deep reinforcement learning; EMI, Electromagnetic interference; ENMs, engineered nanomaterials; FE, Finite element; FEA, Finite element analysis; FFNN, feed-forward neural network; GAN, Generative adversarial network; GBDT, Gradient boosting decision tree; GBRT, Gradient boost regression tree; GCN, Graph convolution network; GMM, Gaussian mixture model; GRNN, Generalized regression neural network; GRU, Gated recurrent unit; H-AgNCs, hairpin-DNA-templated AgNCs; hBN, hexagonal boron nitride; HR, high-resolution; IDP, Instant detection platform; KNN, K-nearest neighbor; LASSO, Least absolute shrinkage and selection operator; LDA, Linear discriminant analysis; LLE, Locally linear embedding; LR, Low-resolution; LSA, Latent semantic analysis; LSM, Log structured merge; LSTM, Long short-term memory; MAE, Mean absolute error; MD, Molecular dynamics; MFs, Metabolic fingerprints; ML, Machine learning; MLANs, Machine learning applications in nanomaterials; ML-PEMST, Machine learning-based property-environment-multi-species toxicity; MNM, Metallic nanomaterial; MOFs, Metal-organic frameworks; MOGAD, Myelin oligodendrocyte glycoprotein-associated disorder; MVR, Multivariate regression; Nano-QSARs, Nano-quantitative structure-activity relationships; NBM, Naive bayes model; NMOSD, Neuromyelitis optica spectrum disorder; NP, Nanoparticle; NPELDI MS, Nanoparticle-enhanced laser desorption/ionization mass spectrometry; NSMP, Non-uniform sequential mobile packing; OSN, Organic solvent nanofiltration; PCA, Principal component analysis; PGB-LTH, Polyethyleneimine/graphene oxide/bentonite clay/layered triple hydroxide; PP, Polypropylene; PPO, Proximal policy optimization; RABs, residual attention blocks; RBFNN, Radial basis function neural network; RC, Residual connection; RCA, Relative contact angle; RFs, Random forests; RMSE, Root mean squared error; RNAB, Residual non-local attention block; RNAN, Residual non-local attention network; RNN, Recurrent neural network; ROIs, Regions of interests; ROM, Reverse osmosis membrane; ROT, Reverse osmosis technology; RP, Relative permeability; RPCA, Robust principal component analysis; RS, Relative selectivity; RSA, Random sequential absorption; RSP, Relative salt passage; R-TNN, Residual connected three-component neural network; RVE, Representative volume element; RWP, Relative water permeability; SAC, Soft actor-critic; SARSA, State-action-reward-state-action; SEM, Scanning electron microscopy; SR, Super-resolution; STEMEELSSI, Scanning transmission electron microscopy-based electron energy loss spectroscopy spectral imaging; SVD, Singular value decomposition; SVM, Support vector machine; SVR, Support vector regression; SWCNTs, single-walled carbon nanotubes; TEM, Transmission electron microscopy; TFN, Thin-film nanocomposite; TFN-ROMs, Thin film nanocomposite reverse osmosis membranes; TRPO, Trust region policy optimization; WP, Water permeability; XGBoost, extreme gradient boosting.

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1. Introduction

With the continuous development of materials science and nanotechnology, nanomaterials, as a new type of material, have attracted widespread attention [1–7]. Nanomaterials possess many unique physical, chemical, and biological properties, such as the size effect, surface effect, and quantum size effect, which make them promising for applications in fields like electronics, energy, environment, and biomedicine [8–11]. However, the design, synthesis, and application of nanomaterials still face many challenges. Traditional experimental methods often require a significant amount of time and resources and are difficult to predict and control the performance of materials. Therefore, a new auxiliary optimization method is needed to further accelerate the development and application of nanomaterials.

Machine learning (ML) is an important branch of artificial intelligence (AI), involving disciplines such as statistics, algorithm optimization, approximation theory, probability theory, and convex analysis, and is key to achieving intelligence [12,13]. The development of ML has gone through four stages: the initial germination period, the preliminary formation period, the rapid development period, and the vigorous growth period. From the original concept of “thinking machines” to today’s deep learning models, ML has gradually evolved into a powerful intelligent technology [14,15]. Fig. 1 depicts the important milestones in the development of ML. ML can extract useful information from a large amount of data and predict future trends. In the field of nanomaterials, ML can be used for predicting material properties, optimizing material structures, and designing material synthesis pathways. With ML, potential nanomaterials can be quickly screened and their performance predicted. This can greatly shorten the research and development cycle of nanomaterials and reduce experimental costs. At the same time, ML can also help people better understand the relationship between the structure and properties of nanomaterials, thus providing guidance for the design of new nanomaterials. In addition, ML can also be combined with other technologies, such as computational materials science and experimental techniques, to form a multidisciplinary research approach. Therefore, the application of ML in the field of nanomaterials is of great significance.

Although many papers have reported satisfactory effects of ML and related research on the preparation and performance of nanomaterials in recent years, to our knowledge, the application of ML in the field of nanomaterials is still rapidly developing, and there is currently a lack of systematic and comprehensive reviews. Therefore, this paper timely summarizes the latest progress in the research of ML applications in

nanomaterials (MLANs) in recent years, fully demonstrating the development and evolution of this field, the current state of research, and research hotspots. It focuses on analyzing the role of ML in promoting the structural design, performance improvement, and application of nanomaterials, which is instructive and valuable for the subsequent optimization of ML methods in the development of high-performance nanomaterials. The research framework of this work is shown in Fig. 2.

2. Principal algorithms and general process of MLANs

Nanomaterials serve as the foundational substances for fabricating high-precision components, machinery, and complex material products. ML, one of the pivotal branches in AI [16], offers strategies for new methods, technologies, and conceptual frameworks in materials science. With the advancement of science and technology, ML has seen rapid growth in its applications within the field of nanomaterials, playing a significant role in driving the development and application of these materials. This section will analyze and discuss various algorithms and application processes of ML in the domain of nanomaterials.

2.1. Principal algorithms

Selecting an appropriate model based on the target problem is the most critical step in nanomaterial research. Different ML algorithms correspond to different scopes of application, and currently, no single algorithm can address all problems. According to the information provided by training samples and the differing feedback mechanisms, the commonly used ML algorithms in the field of nanomaterials can be broadly classified into supervised learning [17], unsupervised learning, reinforcement learning [18], neural networks and deep learning [19], and ensemble learning [20], as illustrated in Fig. 3.

(1) Supervised learning algorithms

Supervised learning algorithms train based on existing labeled data to construct a predictive model that classifies or performs regression predictions on new data. During supervised learning, a pattern (such as a learning model) can be learned or established from the training dataset and then applied to infer new instances. Commonly used supervised learning algorithms include SVMs, decision trees (DTs), GBRTs, and logistic regression, among others. These algorithms are applicable to tasks like material classification and property prediction in the field of nanomaterials.

(2) Unsupervised learning algorithms

Unsupervised learning algorithms process data without labels for

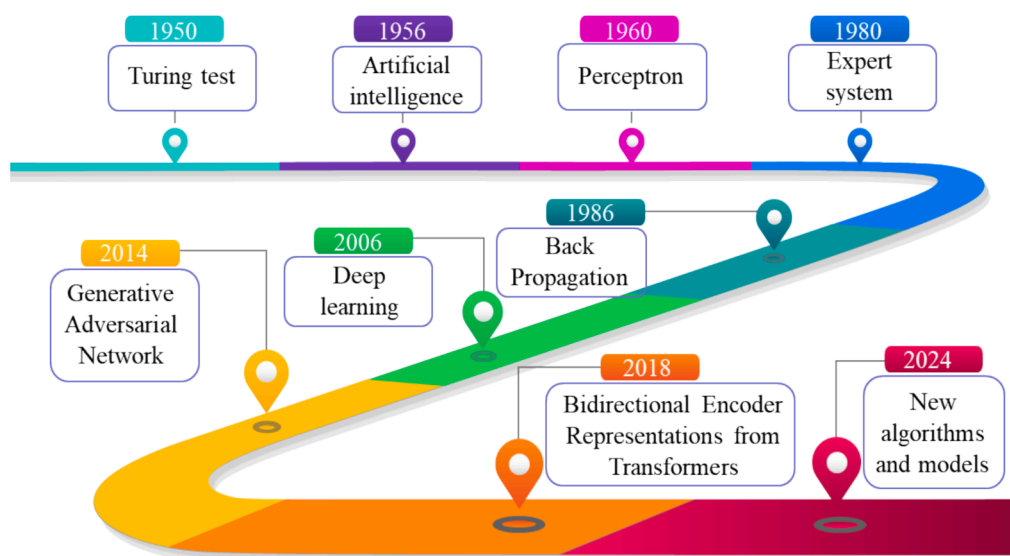


Fig. 1. The development of ML.

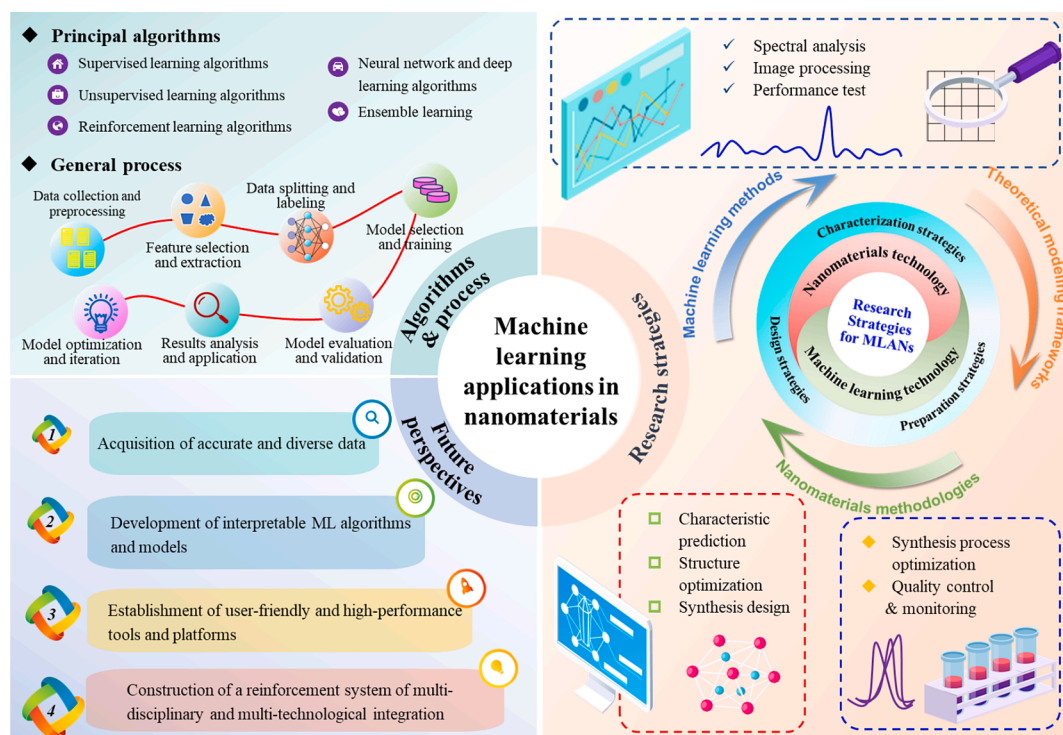


Fig. 2. General frame diagram of MLNs.

clustering, dimensionality reduction, or anomaly detection. These algorithms lack specific target outputs and instead group datasets into different clusters. Notable unsupervised learning algorithms include LSA, K-means clustering, PCA [21], GMM, etc. In the realm of nanomaterials, these algorithms can be employed for similarity analysis, feature extraction, and dimensionality reduction tasks.

(3) Reinforcement learning algorithms

Reinforcement learning is a method where an intelligent agent learns optimal strategies through interaction with an environment. In this context, the agent adjusts its behavior based on feedback (rewards or penalties) from the environment with the goal of maximizing cumulative rewards. Reinforcement learning is highly adaptable and primarily trains through decision-making, where a well-trained algorithm after extensive experience can deliver accurate predictions. Within the field of nanomaterials, reinforcement learning can be applied to decision-making in material design or optimization processes.

(4) Neural network and deep learning algorithms

Neural networks are composed of adaptive units interconnected in a wide and parallel manner, characterized by massive parallel processing, distributed information storage, and excellent self-organizing and self-learning capabilities. Deep learning extends neural network models, leveraging multiple layers of nonlinear transformations for feature learning and pattern recognition. Popular deep learning algorithms include CNNs, RNNs, and deep generative models, among others. In the field of nanomaterials, deep learning algorithms can be utilized for tasks such as image recognition and feature extraction of nanomaterials.

(5) Ensemble learning

Algorithms ensemble learning combines multiple learners to accomplish a learning task, aiming to improve the performance of models (classification, prediction, function estimation, etc.) or reduce the likelihood of choosing inappropriate models. Ensemble learning algorithms are *meta*-algorithms that build and integrate several ML techniques into a single predictive model to achieve effects like variance reduction (bagging), bias reduction (boosting), or improved predictions (stacking). Common ensemble learning algorithms include random forests (RFs), adaptive boost (Adaboost), GBDT, extreme gradient

boosting (XGBOOST), etc. In the field of nanomaterials, ensemble learning algorithms can be applied to tasks such as predicting the properties of nanomaterials and classifying materials.

The structural, morphological, and property characteristics of nanomaterials can be treated as input data, while ML algorithms can learn patterns and rules from these data through training, transforming them into predictive, classifiable, or optimizable models. By using ML algorithms, one can predict the properties of nanomaterials, classify materials, or identify suitable material design or optimization schemes. Moreover, the deep feature learning capability of deep learning algorithms helps to uncover nonlinear relationships between the inherent structure and properties of nanomaterials. Thus, ML algorithms provide an efficient, rapid, and reliable approach to data analysis and modeling in nanomaterial research.

2.2. General process

Compared to traditional homogeneous and single materials, nanomaterials possess greater data volume due to their anisotropic properties, diverse design parameters, and different performance requirements, making the design and optimization process of nanomaterials more complex. Therefore, how to apply ML to nanomaterial research to accelerate the development of nanomaterials and promote their rapid application in the engineering field is an important scientific issue facing nanomaterial research. The workflow of applying ML in the field of nanomaterials can be summarized in Fig. 4.

(1) Data collection and preprocessing

Typically, relevant nanomaterial data is collected through experimental testing and numerical simulation, including structure, morphology, properties, etc. However, these data often suffer from issues such as excessive noise, data redundancy, and data loss. Therefore, data preprocessing is necessary, including cleaning, denoising, outlier removal, etc., to ensure data quality and consistency.

(2) Feature selection and extraction

Based on the characteristics of nanomaterials, the most representative and distinctive features are selected from the raw data, while

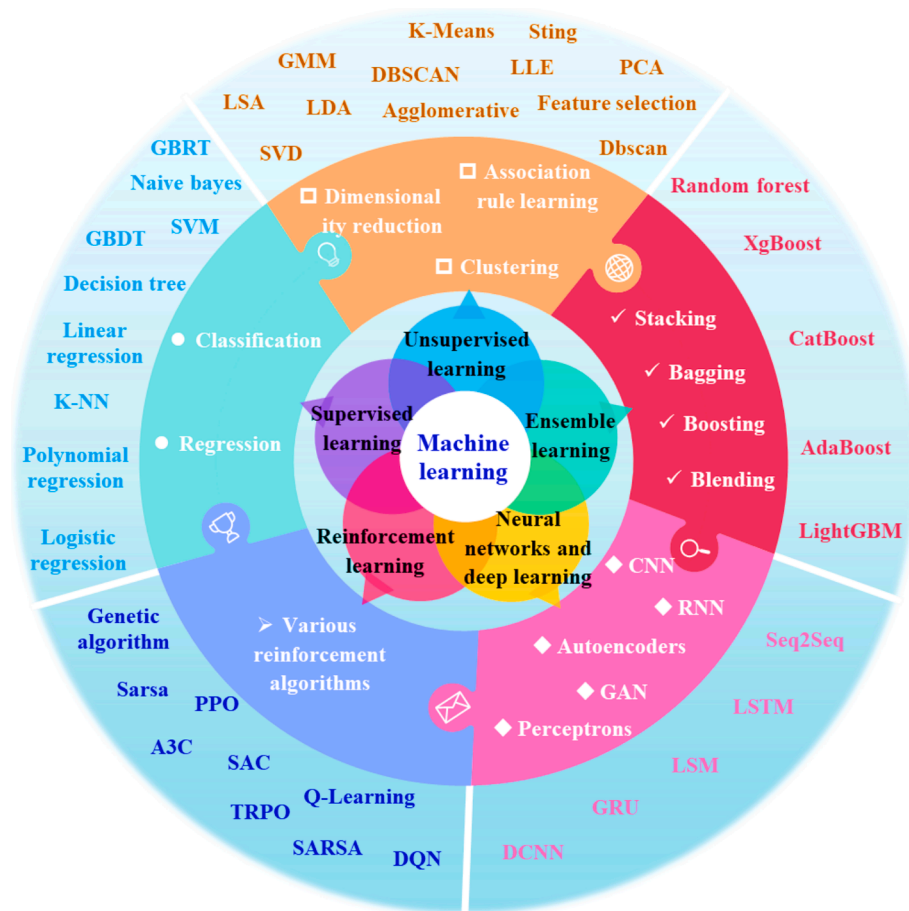


Fig. 3. Potential ML algorithms in the field of nanomaterials. The relevant abbreviations in the figure are shown below: Asynchronous advantage actor-critic (A3C), Convolutional neural network (CNN), Density-based spatial clustering of applications with noise (DBSCAN), Deep convolution neural network (DCNN), Deep Q-learning network (DQN), Generative adversarial network (GAN), Gradient boosting decision tree (GBDT), Gradient boost regression tree (GBRT), Gaussian mixture model (GMM), Gated recurrent unit (GRU), K-nearest neighbors (KNNs), Linear discriminant analysis (LDA), Locally linear embedding (LLE), Latent semantic analysis (LSA), Log structured merge (LSM), Long short-term memory (LSTM), Principle component analysis (PCA), Proximal policy optimization (PPO), Recurrent neural network (RNN), Soft actor-critic (SAC), State-action-reward-state-action (SARSA), Singular value decomposition (SVD), Support vector machine (SVM), Trust region policy optimization (TRPO).

redundant and irrelevant features are removed to reduce feature quantity, improve model performance, and prevent overfitting. Feature selection aims to identify features that contribute significantly to classification or regression, reducing dataset complexity and enhancing model generalization. In addition to selecting appropriate features, feature extraction is also conducted to reduce the dimensionality of the raw data, enhance data representation, and assist ML algorithms in tasks such as classification, clustering, and regression. These processes can be accomplished through physical models, empirical knowledge, or data-driven methods.

(3) Data splitting and labeling

The dataset is divided into a training set and a test set. The training set is used to build models, while the test set is used to evaluate the model's effectiveness. Data splitting methods such as holdout cross-validation, leave-one-out cross-validation, and k-fold cross-validation can be employed [22]. Data labeling involves detailed classification, marking, and annotation of images, videos, audio, text, etc., to assign different labels to data for various ML applications.

(4) Model selection and training

In the field of nanomaterials, complex relationships often exist between given parameters and performance that are difficult to interpret rationally. ML models can fit and extrapolate data directly, mining the intrinsic relationships of nanomaterials. Suitable ML models such as SVM, DTs, RFs, neural networks, etc., are selected based on task

requirements and data characteristics. The model is trained using the training set, and model parameters are tuned to enhance performance and generalization.

(5) Model evaluation and validation

After training the ML model, evaluating the model's effectiveness is crucial. By comparing model output with actual results, the accuracy of the model built using current data and its performance on unseen data can be assessed to select the most accurate model. Additionally, the model's generalization and prediction capabilities are validated using the test set.

(6) Results analysis and application

Interpret and analyze the model's output results, explore the relationship between nanomaterial structure and properties, formulate corresponding application strategies based on the results, and apply them to unknown spaces.

(7) Model optimization and iteration

Based on results analysis and application feedback, optimize and improve the model, including feature engineering, algorithm tuning, etc., to enhance model accuracy and efficiency.

It is important to note that data in the field of nanomaterials often exhibit high dimensionality and complexity. Therefore, adjustments and optimizations should be made in model selection and feature engineering based on practical considerations to improve the effectiveness of ML applications in the field of nanomaterials.

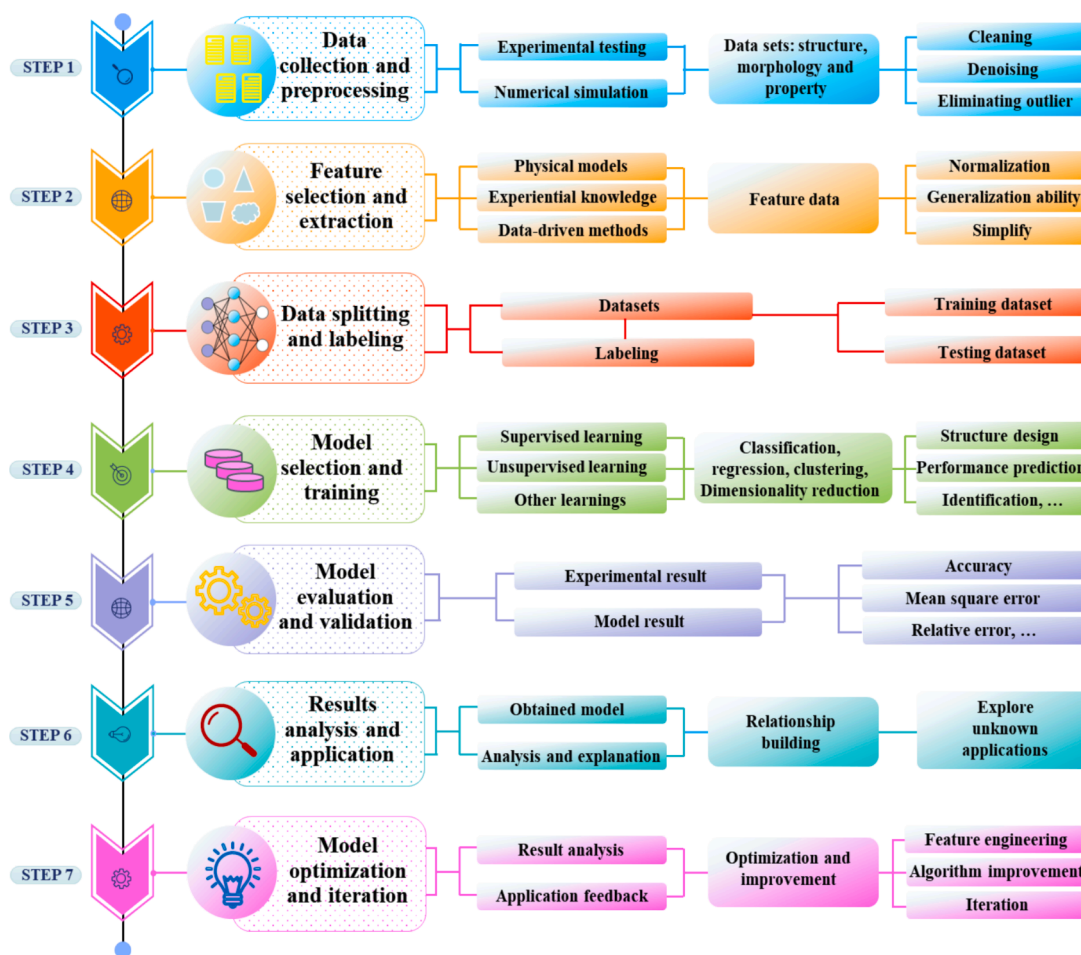


Fig. 4. General process of MLANs.

3. Research strategies of MLANs

In the field of nanomaterials, the combination of ML methods with traditional theoretical and experimental approaches can assist scientists in discovering new materials, rapidly screening materials on a large scale, predicting material properties, accelerating crystal structure searches, and optimization, among others. Fig. 5 illustrates various research strategies aided by ML for nanomaterials.

3.1. ML-assisted design strategies for nanomaterials

The design and application of nanomaterials have become one of the hot topics in current materials science research. Nanomaterials exhibit unique physical, chemical, and biological properties, thus holding broad prospects for applications in areas such as energy storage, sensors, and catalysts. However, due to the size effect, surface effect, and other special properties of nanomaterials, traditional experimental methods and theoretical simulations have certain limitations in nanomaterial design. ML, as a data-driven approach, can effectively address these issues, providing new ideas and methods for the design of nanomaterials. This section will discuss the research strategies of ML in nanomaterial design from three aspects: material property prediction, material structure optimization, and material synthesis pathway design.

3.1.1. Characteristic prediction

Predicting the properties of nanomaterials is an important part of nanomaterial design. Traditional theoretical calculation methods such as density functional theory (DFT) and molecular dynamics (MD) simulations [23,24] can predict some basic properties of materials, but due

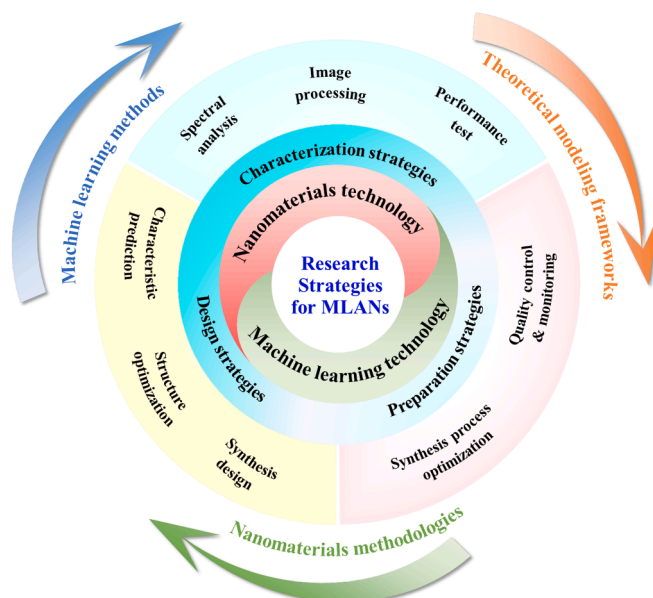


Fig. 5. Various research strategies of MLANs.

to high computational costs and long durations, they struggle to efficiently predict properties of a large number of nanomaterials. ML methods can analyze a vast amount of experimental or computational simulation data to establish predictive models, thereby quickly and

accurately predicting the properties of nanomaterials. These predictions provide important references for the design and application of nanomaterials and help researchers select promising nanomaterials, accelerating the material development process.

Nanoporous materials, characterized by high surface area and porous structures, have significant potential for applications such as adsorption, catalysis, and separation. However, the variety and quantity of nanoporous materials are vast, and discovering materials through

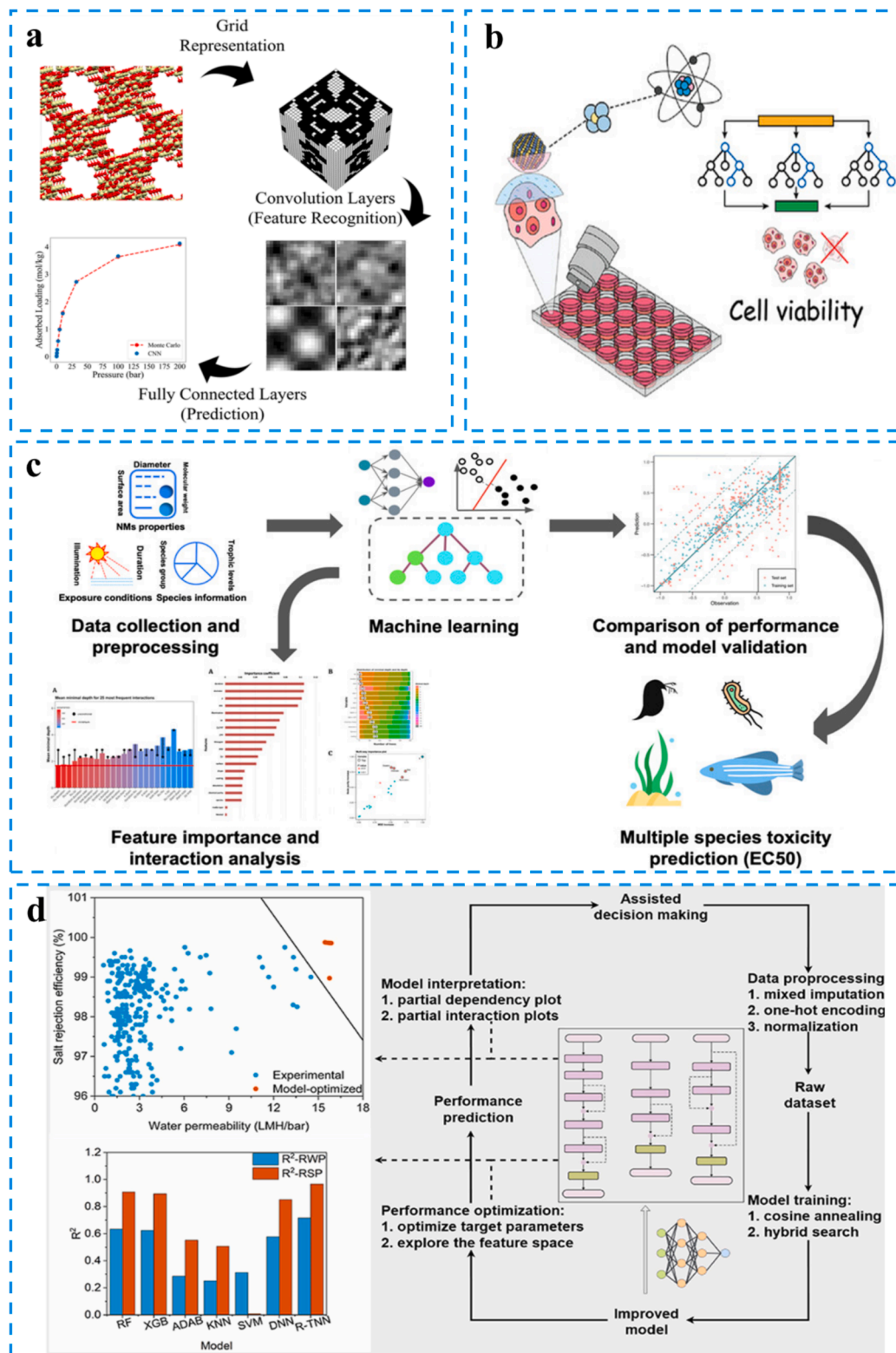


Fig. 6. Design of nanomaterials aided by ML. (a) Adsorption isotherm prediction of nanoporous materials by CNN model [34], (b) Prediction of in vitro cytotoxicity of inorganic nanomaterials based on ML [41], (c) Multi-species toxicity prediction of aquatic environment by ML-PEMST [49], (d) R-TNN model structure, comparison of determination coefficients of various models, and comparison of model performance with existing performance [60]. Acronyms are explained below: RFs, extreme gradient boosting (XGB), adaptive boosting trees (ADAB), KNN, SVM, fully-connected neural network (DNN).

pure trial-and-error or brute-force searching is challenging [25,26], which can lead to human bias, lack of expertise, and laborious manual work. Constructing a ML model to predict material properties is an effective strategy [27,28], such as building a model based on the basic properties of nanoporous materials to predict adsorption characteristics [29,30]. These models enable the correlation of material geometric features with adsorption properties, establishing an intrinsic mechanism of structure and performance [31–33]. The CNN are suitable for more complex material property discovery, and a three-dimensional CNN has been proposed to predict the adsorption performance of nanoporous materials [34], as shown in Fig. 6(a). The self-learning capability of CNNs allows them to be unaffected by manual crafting or human bias, providing highly accurate predictions, and the self-learned features are akin to the structural channels and porous geometric structures.

Despite significant successes in material property prediction with ML assistance [35,36], existing descriptors become insufficient when it comes to nanoparticles, as they lack information about their structure and surface properties. Accurate representation of nanomaterials requires a variety of descriptors for different nanomaterial structures [37,38], for predicting various biological properties [39]. Nanomaterial toxicity is a key parameter in screening biological interactions in each biologically relevant study, which not only strongly depends on the chemical composition and surface modification of the nanomaterials but also on shape, surface area, linear dimensions, size distribution, surface charge, porosity [40]. To prevent excessive experimentation, a model is needed that can pre-assess this parameter. Current models only predict cytotoxicity in binary toxic/non-toxic classifications, with little information on concentration dependence. Based on this, a ML-based method has been proposed to quantitatively predict the in vitro cytotoxicity of inorganic nanomaterials [41], as shown in Fig. 6(b). It achieves feature selection based on correlation to ensure better performance, where hyperparameter grid search allows for the screening of 40 ML models and further optimization of the best model. Atomic-based quantitative nanomaterial descriptors are introduced to provide the model with flexibility, that is, the ability to work beyond the chemical entities presented in the dataset. Feature importance is utilized to find interpretable models with optimal decision-making processes. This model can fully screen for toxicity, eliminating excessive, labor-intensive experiments, enabling the rapid design of nanomaterials for biomedical purposes.

Metallic nanomaterials (MNMs) have seen widespread use [42], which has led to increased discharges into aquatic environments, thereby posing serious toxicity risks to aquatic organisms [43,44]. The complex nature of different MNMs coupled with the complexity of aquatic environments influences the toxic effects of MNMs [45], making experimental assessment of toxicity time-consuming and costly. Predicting toxicity is crucial for a variety of aquatic species, yet it presents substantial challenges [46]. Unlike nano-quantitative structure–activity relationships (nano-QSARs) models [47] that focus on the physicochemical properties of materials, multi-species models [48] can describe distinctive traits across organisms. ML methods offer an alternative to experimental approaches, effectively handling heterogeneous data and identifying relationships among various complex factors. A ML-based property–environment–multi-species toxicity prediction model (ML-PEMST) has been proposed [49] (as shown in Fig. 6(c)), which takes into account the physicochemical properties of MNMs, environmental factors, and diverse biological exposures under different conditions. Research shows that exposure duration, light intensity, primary size, and hydrodynamics are major factors influencing the toxicity of MNMs to aquatic organisms. This model proves effective in predicting multi-species toxicity in aquatic environments and provides guidance for risk assessments of MNMs.

The reverse osmosis technology (ROT) is a critical means for seawater desalination [50], achieved by filtering out salts and impurities from seawater through reverse osmosis membranes (ROMs) to obtain fresh water, thus reducing power consumption and related chemical

requirements [51]. Water permeability (WP) and salt rejection efficiency are vital characteristics of ROMs [52], and the inherent trade-off complexity between these properties poses significant challenges in balancing them [53]. Thin-film nanocomposite membranes represent a typical type of reverse osmosis membranes (TFN-ROMs) [54,55], their performance is conventionally predicted using traditional techniques, but these are more suited for small-scale simple datasets and are limited when dealing with large-scale data. ML, with its strong self-learning ability and excellent generalization capabilities [56], has drawn attention from researchers [57–59]. A residual connected three-component neural network (R-TNN) was designed [60] to understand the relative water permeability (RWP) and relative salt passage (RSP) characteristics of TFN-ROMs, as depicted in Fig. 6(d). The results indicate that despite the high informational complexity and insufficient data samples concerning RWP, R-TNN maintains good predictive power, attributed to its integration of three sub-networks within the model, incorporating residual connections (RC) and selective regularization techniques, granting the entire network powerful learning capabilities, and maintaining robust multi-output functionality. Properties of nanoparticles (NPs), including NP shape, charge, pore size, and NP size, along with the relative water contact angle (RCA), significantly impact RWP and RSP, making these parameters important considerations in the structural design of TFN-ROMs. Furthermore, the R-TNN model can be integrated with other techniques, offering new insights to guide the design of TFN-ROMs; however, issues remain, such as incomplete learning due to insufficient experimental datasets and less satisfactory predictions for RSP.

3.1.2. Structure optimization

Structural optimization of nanomaterials refers to the process of adjusting parameters such as the crystal structure and atomic composition of materials under given conditions to achieve superior performance. Traditional structural optimization methods like genetic algorithms and particle swarm algorithms may have issues such as high computational costs and slow convergence, making it difficult to effectively optimize complex nanomaterials. By learning from a large amount of structure–performance data, ML can build models to predict the relationship between material performance and structure, thereby guiding the structural optimization and helping researchers design more efficient and stable nanomaterials.

Nanocomposites typically consist of nanoparticles, polymers, and small molecules, and their spatial structure is influenced by factors such as polymer chain length, nanoparticle size, organic small molecules, interfacial effects, and environmental conditions [61,62]. Although nanocomposites can self-assemble into functional devices with structure-dependent properties [63], predicting the relationship between their composition and assembled structure is challenging [64], and currently relies mainly on trial and error. ML methods offer a data-driven approach to visualize high-dimensional spaces and predict outcomes for complex systems [65]. A study has been conducted using a dataset of micrographic images of nanocomposite films to train a ML model [66], which analyzes the correlation between input and output parameters and establishes a data analysis pipeline. The resulting ML model can be effectively used to predict the structure of new nanocomposite mixtures, demonstrating the potential of ML technology in improving the efficiency of nanocomposite device design.

Piezoelectric oxides [67,68] possess high piezoelectricity and excellent electromechanical coupling efficiency, while piezoelectric polymers [69] offer mechanical flexibility and biocompatibility. Piezoelectric nanocomposites incorporating oxide fillers into a polymer matrix have garnered widespread attention in the fields of flexible electronics and energy harvesting devices. The material properties and filler structure are key to achieving the desired functions. By integrating the advantages of high-throughput phase field methods with ML [70], Li et al. [71] studied the impact of the morphology and spatial orientation of oxide fillers on the performance of nanocomposites through high-

throughput phase field simulations and established a ML model with piezoelectric coefficients, dielectric constants, and stiffness as inputs. Compared to other geometric structures, nanopillar fillers with the largest aspect ratio provide the composite with better piezoelectric response and electromechanical coupling coefficients. The combination of phase field simulation and ML technology can provide material design strategies for high-performance wearable electronic products.

The mechanical properties of materials are highly dependent on their microstructural characteristics, including the shape, dispersion, and orientation of inclusions and/or voids [72,73]. The inconsistent nature of inclusion aggregation and the irregularity in shape, size, and orientation of inclusions and/or voids hinder the local control of the arrangement and dispersion of inclusions and/or voids, resulting in a variety of non-uniform and complex microstructures [74]. Characterizing the process-microstructure-property relationships of material systems through experiments is challenging and expensive. Finite element (FE) analysis models provide an effective method for virtual experiments on different material systems during the design process [75], but determining the specific microstructure of heterogeneous and composite materials is difficult due to irregular microstructural characteristics. Statistical microstructural evaluation provides a way to characterize microstructural features [76], such as representative volume elements (RVEs). Algorithms like image-based geometry reconstruction [77], MD [78], random sequential absorption (RSA) [79] can characterize the microstructure of various material systems, but these algorithms cannot generate three-dimensional microstructures with high discontinuity and other complex microstructural features with different inclusion shapes and random sizes. To generate RVEs with different microstructural characteristics, a multifaceted, adjustable non-uniform sequential mobile packing (NSMP) algorithm has been developed [80] for generating various possible microstructures (porous inhomogeneous systems and

composite material systems). Fig. 7 is a general flowchart of the NSMP algorithm, with input variables including the size of the inclusions, the size of the initial cubic area, the minimum allowable distance between two inclusions, the size of the RVE, the shape of the inclusions, and microstructural characteristics. Twelve microstructures were generated, and statistical finite element micromechanics modeling was used to calculate the effective properties of the microstructures. The ability of the developed algorithm to generate representative microstructures based on the degree of heterogeneity contained in the RVEs was assessed. In 3D printed nanocomposites, aligned and misoriented rod inclusions are the most representative microstructures. Contour plots of normalized effective stiffness and normalized mean hydrostatic stress can be used to compare different microstructures of RVEs. RVEs with aligned rod inclusions have the highest modulus and the lowest mean hydrostatic stress in the matrix, especially at higher inclusion volume fractions, representing the best performance in the microstructure. The NSMP algorithm can be used for further microstructural optimization and the assessment of regenerated or porous heterogeneous and/or composite materials using ML, and the generated profiles can serve as general design guidance.

3.1.3. Synthesis route design

The design of synthetic pathways for nanomaterials typically refers to the precise synthesis of such materials by adjusting synthesis parameters under given conditions. Traditional methods for designing these pathways rely heavily on experimental trials and accumulated experience, which often suffer from lengthy synthesis periods, high costs, and inefficiency in the design of complex nanomaterials. Researchers have integrated ML techniques to develop more efficient chemical synthesis routes, thereby accelerating the design process for novel nanostructures and optimizing existing materials [81]. ML

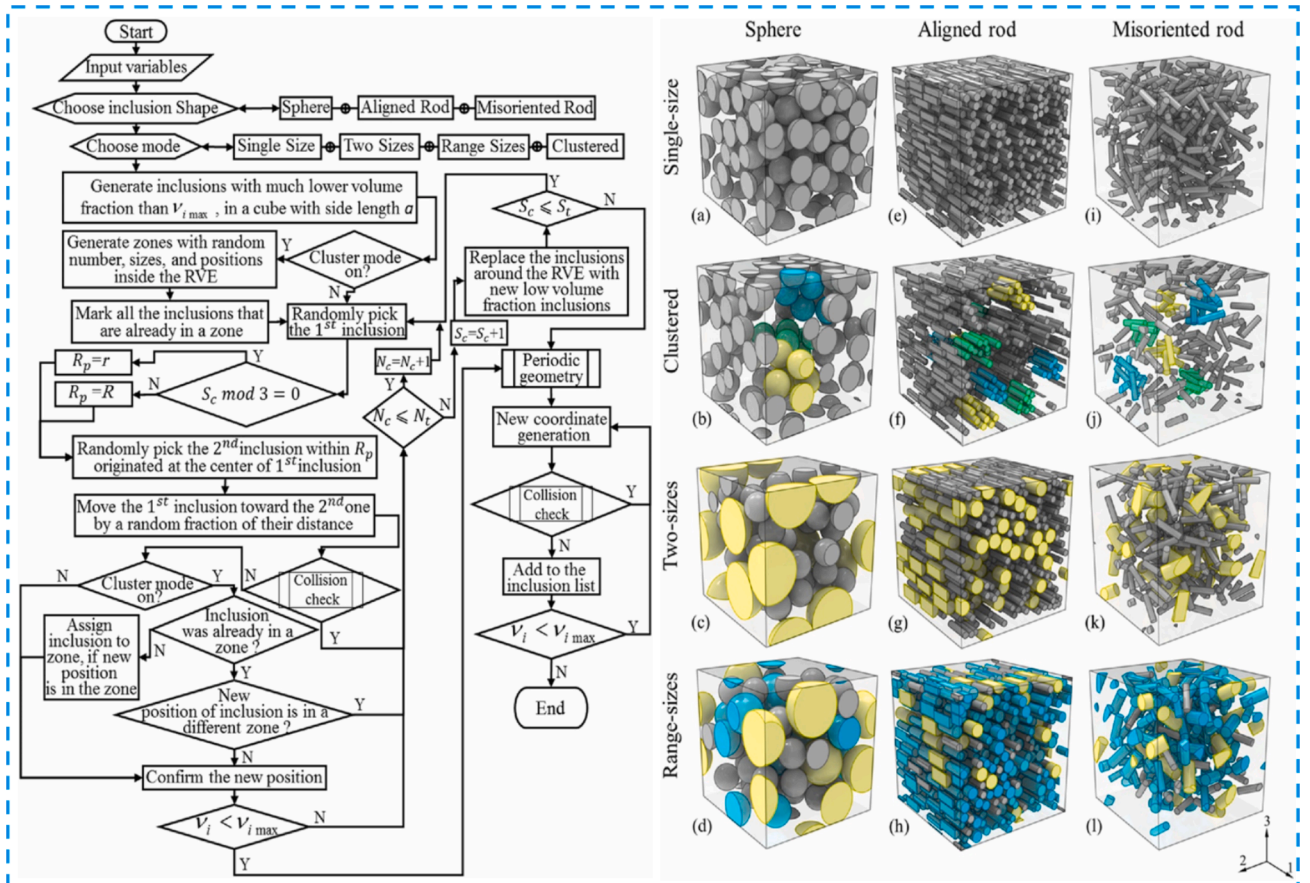


Fig. 7. General flowchart of the NSMP algorithm and the 12 microstructures generated by the NSMP algorithm [80].

technology can be trained to learn from past experiments and guide future synthetic pathway design processes. This data-driven optimization approach not only increases the speed of material design but also effectively saves time and money, while reducing various forms of waste generated during material synthesis and optimization [82].

Silver nanoclusters (AgNCs), distinct from silver nanoparticles, exhibit molecular-like properties, such as fluorescence [83]. To address the irreversible aggregation issue of AgNCs in solution, biomolecules are often employed as templates for stabilizing AgNCs [84], with DNA being a prominent example [85]. DNA-stabilized AgNCs are widely used as fluorescent probes due to their structural stability and tunable fluorescence [86,87]. The DNA sequence [88], structure, and experimental conditions [89] play crucial roles in determining the fluorescence properties of AgNCs, yet their exact relationships remain unclear. ML technologies offer a means to reveal the relationship between DNA sequences and the fluorescence characteristics of AgNCs [90]. Inspired by neurons in biological neural networks that selectively store and transmit sequential data information [91], RNNs can perform similar functions. A deep learning protocol based on a RNN algorithm has been proposed [92] to predict the fluorescence properties of hairpin-DNA-templated

AgNCs (H-AgNCs). The workflow is shown in Fig. 8(a), including H-AgNC synthesis, fluorescence spectral acquisition, statistical data analysis, model training, model evaluation, and experimental testing. By taking stem sequences and synthesis parameters as input features and the fluorescence properties of H-AgNCs as output labels, the RNN model is trained to reveal the impact of input parameters on fluorescence. In order to study the generalization capability of the trained model, 10,000 randomly generated stem sequences of DNAs were synthesized computationally and incorporated into the training model. Results show that the RNN model accurately identifies the fluorescence properties of H-AgNCs with an accuracy exceeding 80%, indicating that the trained RNN model possesses good generalization capabilities. This strategy provides a powerful tool for designing H-AgNCs with desired fluorescence properties and offers a new avenue for the development of DNA-based fluorescent probes.

Carbon dots (CDs), as zero-dimensional carbon nanomaterials, are characterized by simple preparation, high stability, and good water solubility, making them significant in bioimaging and biosensing applications [93]. However, the controlled synthesis of CDs remains immature [94,95] due to factors such as precursor types and amounts,

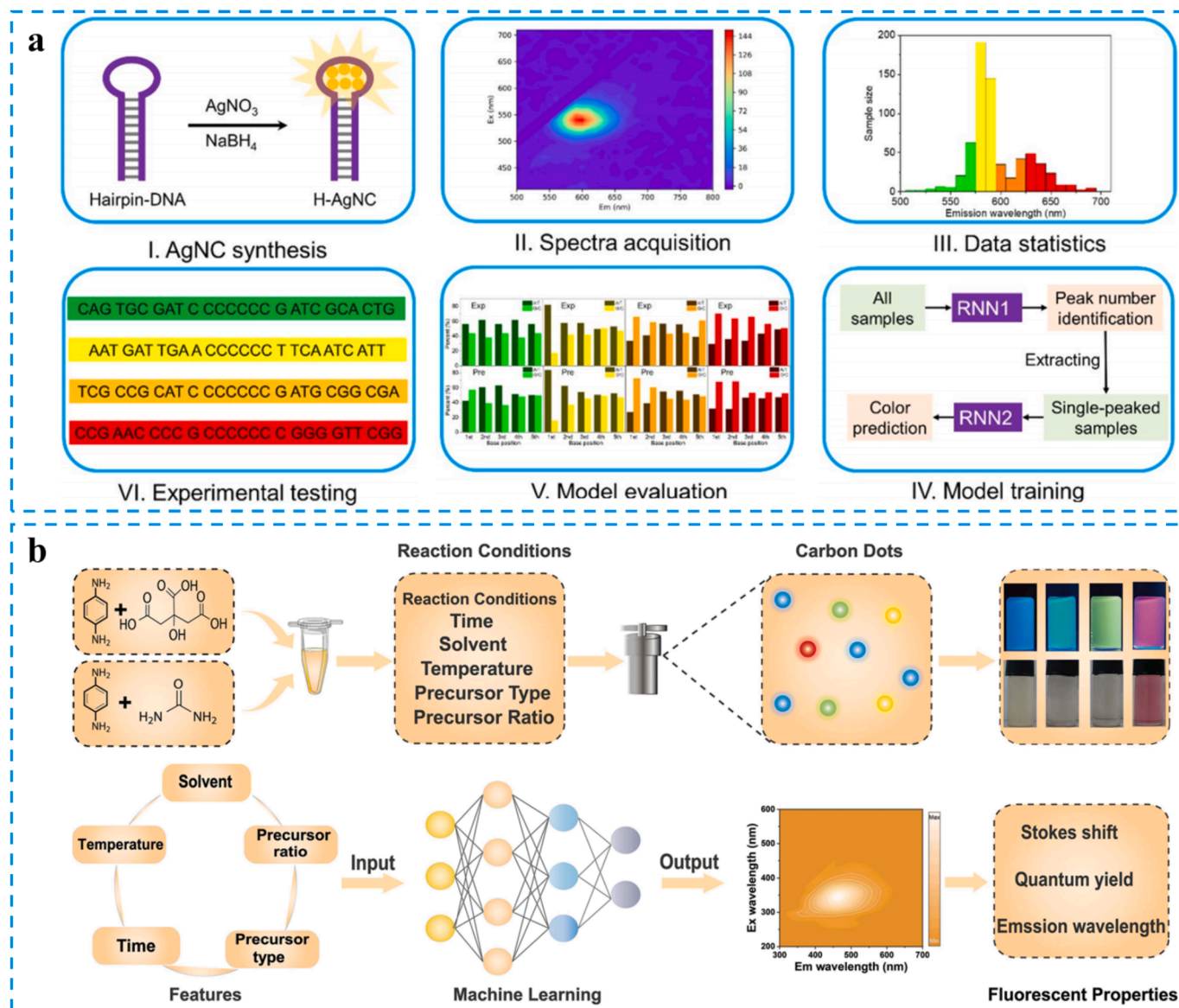


Fig. 8. ML for the design of nanomaterial synthesis pathways. (a) Workflow of RNN algorithm predicting the photoluminescent properties of H-AgNCs [92], (b) Synthesis of multicolor CDs and their ML-assisted workflow [100].

reaction times, temperatures, and solvent types [96]. Traditional trial-and-error methods based on human expertise dominate CD synthesis, but these approaches are inevitably influenced by subjective human factors, hindering the production of CDs with ideal characteristics. ML can provide effective learning models based on extensive data analysis across various applications, capturing hidden information from large-scale experimental and computational simulation datasets [97,98]. Various ML strategies have been applied to CD research, including XGBoost models and DCNNs [99]. Despite this, CD synthesis data often suffers from limited and ambiguous information, complicating the exploration of potential correlations between synthesis conditions and photoluminescence properties. Chen et al. prepared multi-color fluorescent CDs with excellent photoluminescence performance using hydrothermal methods and achieved controllable synthesis using ML [100], as depicted in Fig. 8(b). They initially synthesized 270 varieties of multi-color CDs and compiled a dataset. After model training and optimization, a RF model could accurately predict the photoluminescence characteristics of multi-color CDs. By implementing standardized data collection, employing diverse experimental conditions, utilizing stratified sampling techniques, and combining data augmentation and cross-validation methods, the impact of human bias on the CD synthesis dataset was effectively mitigated. This demonstrated that both reaction parameters and solvent properties have different impacts on the fluorescence performance of CDs, enabling intelligent optimization of synthetic pathways. Moreover, leveraging the rich color information from ML-screened CDs, they were applied in high-capacity color information encryption. Utilizing ML-assisted strategies to guide the synthesis of multi-color CDs and investigate their photoluminescence properties presents a pathway toward intelligent synthesis routes and performance enhancement for CDs.

During the growth process of hexagonal boron nitride (hBN), there are complex interactions and synergistic effects between different growth parameters, such as temperature, gas flow rate, and precursor concentration, and their individual and combined changes have a complex impact on its growth. Traditional data analysis methods often struggle to handle and interpret high-dimensional datasets under multi-parameter variations, especially in complex systems like material growth that involve a large number of variables and potential nonlinear relationships. To improve the synthesis quality of hBN, optimize the growth process, and achieve better control, it is necessary to deeply understand how each growth parameter affects the growth mechanism of hBN. Machine learning is employed to reveal the dependency between the growth dynamics and growth parameters of hBN during the chemical vapor deposition process [101]. By varying growth parameters in the deposition process, such as temperature, hydrogen gas flow rate, and the flow rate of the precursor (borazine), the nucleation density and flake size of hBN are controlled. Utilizing a Gaussian Process machine learning model, the covariance matrix between multi-dimensional input data is analyzed, thereby predicting the mean and variance functions during the hBN growth process. This reveals the correlations between different growth parameters and identifies growth windows to achieve larger flake sizes. This study not only enhances the understanding of the hBN growth mechanism but also provides a new analytical method for the systematic understanding of the growth mechanisms of other two-dimensional materials, helping to optimize the synthesis process and improve the quality and repeatability of the materials.

In summary, ML has significant application value in the design of nanomaterials. By analyzing three aspects: material property prediction, material structure optimization, and material synthesis pathway design, we can find that ML methods can construct models to predict the properties, structure, and synthesis pathways of materials by learning from a large amount of data. This provides new ideas and methods for the design and application of nanomaterials. Therefore, the application prospects of ML in nanomaterial design are broad and will bring new opportunities and challenges to the research and application of nanomaterials.

3.2. ML-assisted characterization strategies for nanomaterials

The application of ML in the characterization of nanomaterials is a widely concerned research field. Through ML methods, spectral data analysis, microscopic image processing, and material performance testing of nanomaterials can be performed, thereby achieving rapid and accurate characterization of nanomaterial properties. In this section, we will analyze three aspects: spectral data analysis, microscopic image processing, and material performance testing, to explore the research strategies of ML in the characterization of nanomaterials.

3.2.1. Spectroscopic analysis

Spectroscopic analysis is a commonly used material characterization technique. The spectral data of nanomaterials include absorption spectra, photoluminescence spectra, Raman spectra, etc., which reflect the optical characteristics of the materials. Traditional spectroscopic data analysis methods rely on professional knowledge and experience for interpretation and face issues such as large amounts of data and complex information, making it difficult to comprehensively and accurately evaluate the optical characteristics of nanomaterials. ML methods analyze the absorption spectra of nanomaterials, predicting the material's band structure, optical bandgap, and other optical properties, providing an important reference for the optical applications of materials. At the same time, by learning from the photoluminescence spectra, Raman spectra, and other data of nanomaterials, predicting the material's photoluminescent performance, Raman characteristics, etc., provides new ideas for the analysis of nanomaterial properties.

Intracellular vesicular transport plays an important role in the body, especially in the uptake, transport, and degradation of engineered nanomaterials (ENMs). ENMs enter cells through active endocytosis and are transported in the endosomal pathway, eventually accumulating in lysosomal vesicles [102]. This process involves various biological pathways, including the regulation of cellular functions such as autophagy, nutrient degradation, and the output of metabolic products. Research on the uptake [103] and spatiotemporal localization of ENMs has made certain progress, and high-spectral microscopy [104] and confocal Raman imaging [105] provide reliable intracellular data that can characterize the complex interactions of ENMs in the biological environment. However, the understanding of the impact of ENMs on the intracellular transport process and how these processes affect intracellular pathways is still limited. Research on the transport and processing of ENMs within cells is crucial for assessing their impact on key cellular functions. Single-stranded DNA can disperse individual single-walled carbon nanotubes (SWCNTs) into a stable colloid, maintaining biocompatibility and surface functionalization [106]. A non-covalent complex formed by SWCNTs and single-stranded DNA (DNA-SWCNTs) has been developed [107], and this formed ENM is used as a signal transduction element to track and report the processing of nanomaterials within cells. To characterize the dynamic behavior of nanomaterials within cells, an artificial neural network (ANN) algorithm was used to predict and classify different types of endosomes containing DNA-SWCNTs, as shown in Fig. 9(a). The Raman spectral data of DNA-SWCNTs were used as the training dataset to train the ANN model, enabling it to identify and distinguish early endosomes, late endosomes, and lysosomes. At the same time, PCA was used as a preprocessing step to reduce the dimensionality of the spectral data and prevent overfitting. The final number of spectra input into the ANN model was reduced to 44 principal components, which accounted for 95% of the variation in the entire training dataset. The prediction results of the well-trained ANN model can effectively classify regions of interests (ROIs) into three different organelle types. This approach, which uses DNA-SWCNTs as internal reporting molecules combined with ML algorithms, provides a way to effectively predict and understand the mechanisms of intracellular transport and processing of nanomaterials.

The CNNs are one of the most commonly used deep learning methods for visual image analysis and classification [108]. However, the

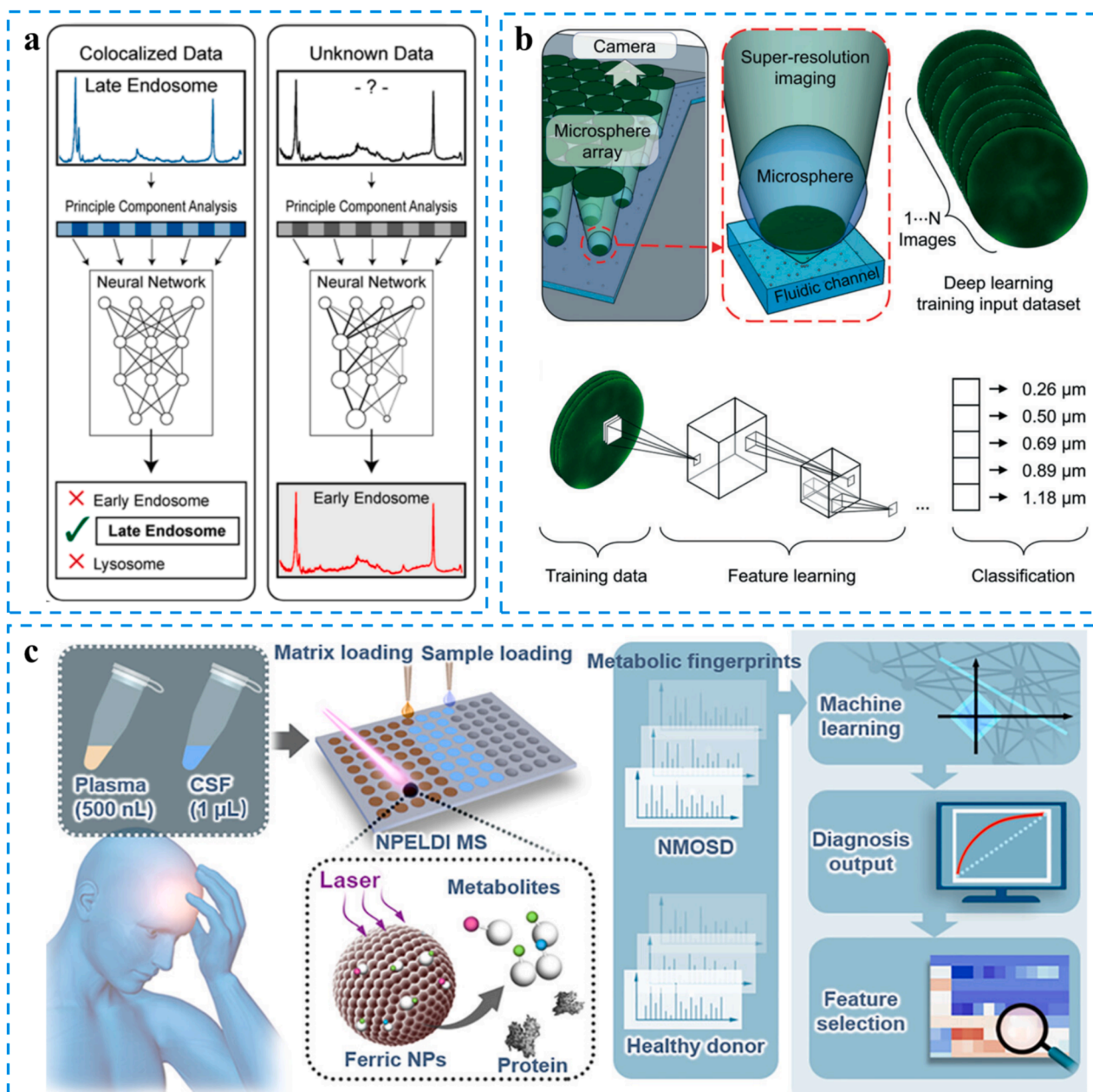


Fig. 9. Spectral data analysis of nanomaterials by ML. (a) ANN processing training and application process [107], (b) Recognition and classification process of intelligent nanoscope [111], (c) Metabolic fingerprint analysis and NMOSD screening process by ML [115].

identification and classification of nanomaterials are still limited by factors such as image resolution, field of view, and processing time. While traditional optical microscopes are widely used in laboratories, their limitations in resolution and field of view make them unsuitable for identifying nanoscale objects. Microsphere technology has been shown to be an effective strategy [109] for overcoming the diffraction limit of optical microscopes by increasing the field of view [110] and reducing sampling time. An imaging platform called “intelligent nanoscope” has been developed [111], as displayed in Fig. 9(b). This platform combines DCNNs with microsphere array imaging technology for rapid identification and classification of nanomaterials. This technology overcomes the limitations of resolution, field of view, and processing time encountered by traditional optical microscopes when identifying and classifying nanoscale objects. Training data sets are obtained through microsphere arrays and traditional imaging methods (bare imaging). A deep Convolutional Neural Network based on AlexNet is trained on this data and applied to the identification and classification of nanomaterials

of different sizes. Results show that using microsphere array imaging methods can achieve higher classification accuracy with smaller data sets compared to imaging methods without microsphere arrays. Through microsphere array imaging methods, the classification accuracy of bacteria can reach 92%.

Neuromyelitis optica spectrum disorder (NMOSD) is an autoimmune antibody-mediated inflammatory disease of the central nervous system characterized by high recurrence rates and poor prognosis [112]. Patients with NMOSD exhibit metabolic disturbances, but there is currently a lack of effective tools to detect the metabolic phenotype of NMOSD. Traditional antibody detection methods suffer from high false-negative rates and long diagnostic intervals [113]. Timely screening and accurate diagnosis are crucial for improving patients’ quality of life [114]. A rapid detection method based on nanoparticle-enhanced laser desorption/ionization mass spectrometry (NP/ELDI MS) has been proposed [115]. This method can obtain metabolic fingerprints (MFs) from trace plasma and cerebrospinal fluid samples within 30 s, as illustrated

in Fig. 9(c). By analyzing these metabolic fingerprints using ML, timely screening for NMOSD has been achieved. A binary diagnostic classifier based on the least absolute shrinkage and selection operator (LASSO) algorithm was constructed to distinguish between NMOSD patients and healthy donors. The LASSO algorithm helps extract significant metabolic features from high-dimensional data sets for model classification. Additionally, an integrated panel was constructed, which showed higher accuracy and specificity in differentiating NMOSD from myelin oligodendrocyte glycoprotein-associated disorder (MOGAD) patients

compared to diagnostic decisions made by 10 clinical experts. This method's superiority lies in its ability to provide high-resolution metabolic fingerprints while maintaining fast detection speeds and low costs. The development of this technology provides an advanced nanomaterial tool for timely screening and differential diagnosis of NMOSD, especially in regions where antibody testing is limited.

Furthermore, ML algorithms have also been applied to predict the light transmittance performance of carbon nanotube (CNT)-loaded polymer nanocomposite membranes [116] and analyze spectroscopic

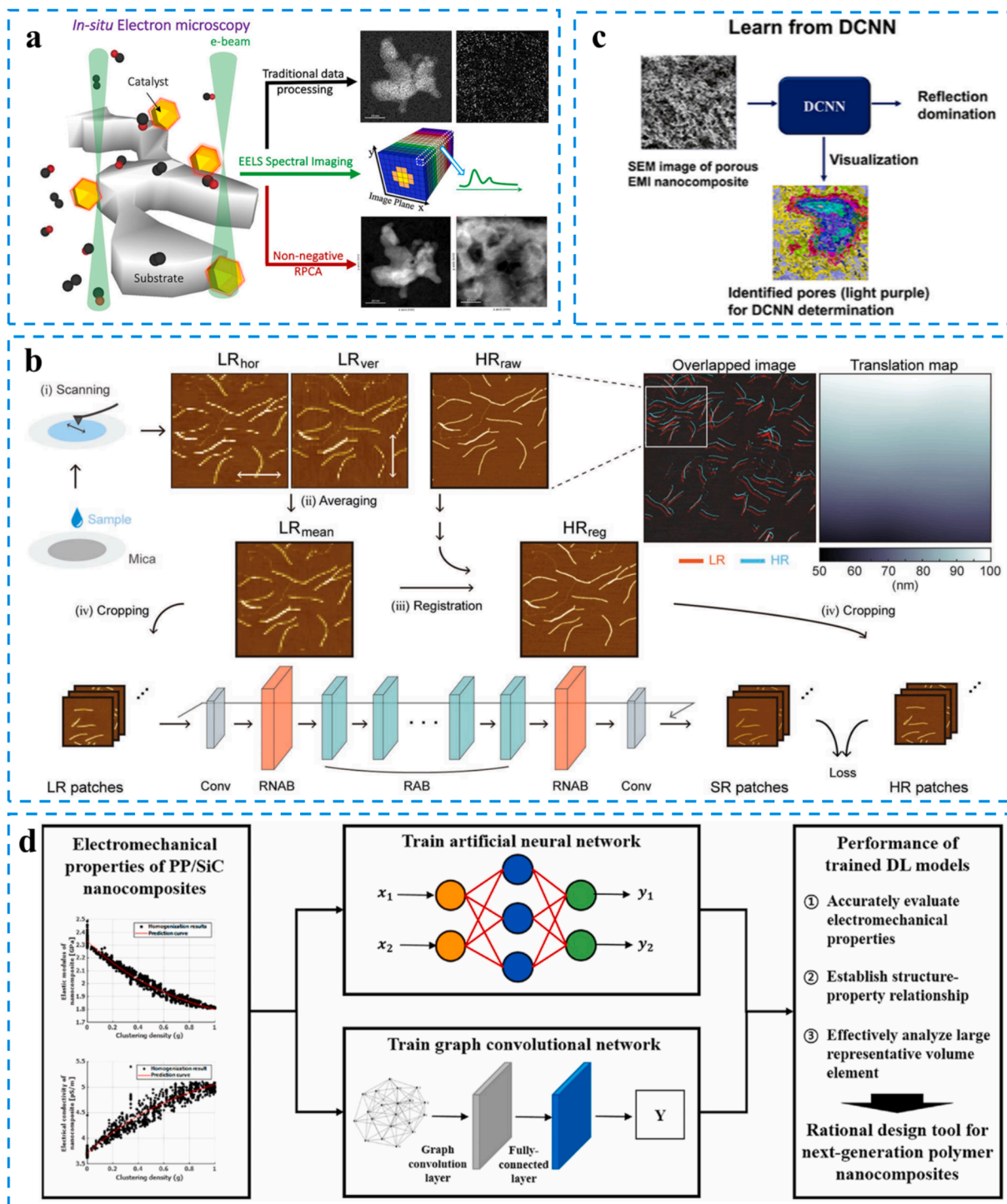


Fig. 10. Characterization of nanomaterials using different ML algorithms. (a) Analysis of microscopic images using ML methods based on RPCA [124], (b) Acceleration of AFM image characterization process using a deep learning-based SR model [128], (c) Visualization of DCNN [133], (d) Process of deep learning-assisted evaluation of the electromechanical properties of PP/SiC nanocomposites [141].

ellipsometry data [117]. The former developed feed-forward neural network (FFNN), generalized regression neural network (GRNN), radial basis function neural network (RBFNN), and multivariate regression (MVR) models, optimizing the performance of each model by adjusting network parameters. The latter used ANN and support vector regression (SVR) to calculate the optical parameters of PVP:ZnTiO₃ nanocomposites. Therefore, ML technology has significant potential for data analysis in nanomaterial science, providing fast and accurate analysis results that contribute to a deeper understanding of nanomaterial properties.

3.2.2. Image processing

The microscopic images of nanomaterials mainly encompass transmission electron microscopy (TEM) and scanning electron microscopy (SEM) [118], which reflect the material's characteristics at the microstructural level. Conventional microscopic image processing methods rely heavily on professional knowledge and experience for image analysis and feature extraction [119], facing challenges such as image noise, segmentation difficulties, and thus struggle to provide comprehensive and accurate assessments of nanomaterials' microstructure. Leveraging a large volume of microscopic images, ML constructs models to automate the analysis and feature extraction of nanomaterials' microscopic images [120], facilitating learning and recognition, thereby enabling rapid and precise prediction of the nanomaterials' microstructural attributes.

Scanning transmission electron microscopy-based electron energy loss spectroscopy spectral imaging (STEMEELSSI) is widely used in materials research [121–123], however, it faces challenges in extracting information from noisy data and overlapping edges. Commonly used EELS spectral imaging analyses lack the ability to isolate noise and eliminate overlapping edges, thereby limiting resolution and signal-to-noise ratio. A ML method based on non-negative robust principal component analysis (RPCA) has been introduced [124], which improves spatial-temporal resolution, signal-to-noise ratio, and discerns subtle differences, as indicated in Fig. 10(a). This ML algorithm has been applied to 13 nanomaterial systems, demonstrating its effectiveness in enhancing image quality compared to traditional methods.

Atomic Force Microscopy (AFM) is widely used for studying static [125] and dynamic [126] properties of various material systems due to its ability to capture high-resolution (HR) images without staining any material. Nonetheless, HR AFM imaging is often hampered by low scan throughput because it requires point-by-point scanning at a constant speed over specific areas, which usually restricts its applications. ML algorithms simplify and automate post-processing analysis of AFM images [127], however, these analyses typically necessitate vast quantities of HR images to train data-driven models for achieving high accuracy and statistical significance. To improve AFM imaging efficiency, high-speed AFM and advanced scanning algorithms have been developed, albeit often requiring expensive equipment and specialized expertise to generate high-quality images. Kim et al. proposed a deep learning-based AFM image super-resolution (SR) method [128] to expedite the characterization process at the nanoscale, as illustrated in Fig. 10(b). Non-contact mode AFM was employed to acquire low-resolution (LR) and HR images of the same imaging region. To minimize image distortion caused by sample movement and deformation, orthogonal scans were performed to generate an average LR image, which was then precisely aligned with the HR image using iterative elastic registration techniques. A residual non-local attention network (RNAN) served as the base SR model, efficiently utilizing the self-similarity within the images through residual attention blocks (RABs) and residual non-local attention blocks (RNABs), enabling successful training of the SR model with only a small set of image pairs. This tailored model could be efficiently fine-tuned for specific target structures via transfer learning, which is more beneficial in practice than building a generic SR model applicable to various target structures. To evaluate the SR model's performance, it was applied to AFM imaging of structured DNA assemblies, where it

successfully reproduced the fine structural details from LR images onto SR images, achieving at least a tenfold increase in measurement speed. Analyses using SR images were consistent with those obtained from HR images, proving the high accuracy and efficiency of the SR model in statistical analysis of AFM samples.

In electromagnetic interference (EMI) shielding materials, constructing porous structures is crucial for reducing electromagnetic wave reflection and secondary pollution [129–131]. Currently, the absence of direct analysis methods for porous structures in EMI shielding materials hampers their advancement. Additionally, the lack of explainability in DCNNs constrains their application in materials science [132]. A method combining DCNN visualization with experiments has been put forward to study porous EMI nanocomposite materials [133], as presented in Fig. 10(c). Improved residual network (ResNet) was trained on SEM images of EMI shielding CNT/PVDF composite samples with varying pore content and filler levels to determine the material's shielding properties. Eigen-CAM visualization indicated that the number and depth of pores influence the shielding mechanism. This visualization technique holds guiding significance for nanomaterial mechanism research and could become a useful tool for characterizing porous structures. Furthermore, ML can classify microscope images, furthering the broad application of nanomaterials. High-throughput SEM combined with neural networks serves as an effective way to determine the morphological handedness of chiral nanoparticles [134,135]. Groschner et al. [136] used label error robust deep learning to classify the handedness of chiral nanomaterials and evaluated minimization of classification errors when automatically labeling training data in SEM datasets. The effects of label error rate and training methods on the classification error of the neural network were analyzed on both ideal and real-world datasets. Three training methods considered were pretraining, co-teaching, and a novel two-step method. Results showed that the co-teaching method performs best in practical applications. Similarly, TEM images of nanostructured materials with different chemical compositions have also been classified using convolutional neural networks [137].

3.2.3. Performance test

Performance testing of nanomaterials includes mechanical properties, thermal properties, electrical properties, etc. These tests reflect the characteristics of materials in terms of mechanics, thermodynamics, and electricity. General material performance testing methods rely on specialized equipment and complex operations, and there are issues such as long testing cycles and high costs, making it difficult to quickly and accurately evaluate the performance of nanomaterials. However, ML methods can learn from a large amount of performance test data, build models to predict the mechanical, thermal, and electrical properties of materials, and achieve rapid and accurate assessment of nanomaterial performance. For example, ML methods can learn from the thermal and electrical properties of nanomaterials, predict the thermal and electrical conductivity of materials, providing new ideas and methods for the analysis of thermal and electrical characteristics of nanomaterials. This can then establish a relationship model between material performance and structure, guiding the design and optimization of nanomaterials.

Despite the potential of ML technology in material design, there are still challenges in predicting the physical properties of polymer nanocomposites, especially when considering the distribution and aggregation state of nanoparticles. Traditional models based on agglomeration density and multivariate regression models have limitations in evaluating the electromagnetic properties of polymer-based composites and cannot accurately reflect the impact of nanoparticle distribution on performance. The physical properties of polymer nanocomposites are adjusted by reinforcing nanofillers [138,139]. The size, distribution, shape, and inherent properties of nanoparticles are key factors affecting the macroscopic performance of nanocomposites. Some multi-scale frameworks for the impact of polymer nanocomposites have been

proposed [140], but existing evaluation methods find it difficult to accurately capture the relationship between structure and performance. To address the above issues, Baek et al. [141] used two deep learning techniques, ANN and graph convolution network (GCN), to establish the complex relationship between the microstructure and macroscopic performance of polypropylene (PP)-based nanocomposites, as shown in Fig. 10(d). The ANN model considers various input features, including the number of nanoparticles, agglomeration state, position, and distribution, while the GCN model transforms the microstructure of the material into graph data, using nodes and edges to describe the interaction and distribution of nanoparticles. By comparing the predictive performance of traditional clustering density models, multivariate regression models, and deep learning models, it was found that deep learning models have significant advantages in evaluating the performance of nanocomposites. The GCN model can learn the impact of the distribution and aggregation of nanoparticles in the polypropylene matrix on material performance, which is difficult for traditional physical models to capture. By combining with multi-scale analysis results, the GCN model not only improves the evaluation performance but also provides performance predictions for large RVEs without losing accuracy, helping

researchers optimize the structure of nanocomposites to achieve better performance. The good predictive results of the GCN model, especially for the physical properties of polymer nanocomposites with non-uniform distribution and dispersed nanoparticles, make it a viable alternative to finite element analysis (FEA).

Dielectric materials play a crucial role in modern electrical and electronic energy systems [142,143]. Energy density and charge-discharge efficiency are two key parameters for evaluating the performance of dielectric materials [144,145], where high energy density and high breakdown strength are ideal characteristics. The relatively low electric displacement of polymer dielectric materials limits their charge storage capacity, resulting in suboptimal energy density. Constructing composite materials by introducing high dielectric constant inorganic nanofillers into polymer matrices can enhance energy density [146,147]. However, the impact and mechanisms of different nanofiller morphologies and orientations on dielectric performance are still not well understood [148]. The energy density range resulting from different types of polymers and nanofillers remains limited, and under high electric fields or temperatures, charge-discharge efficiency may significantly decrease due to increased conduction losses. To address

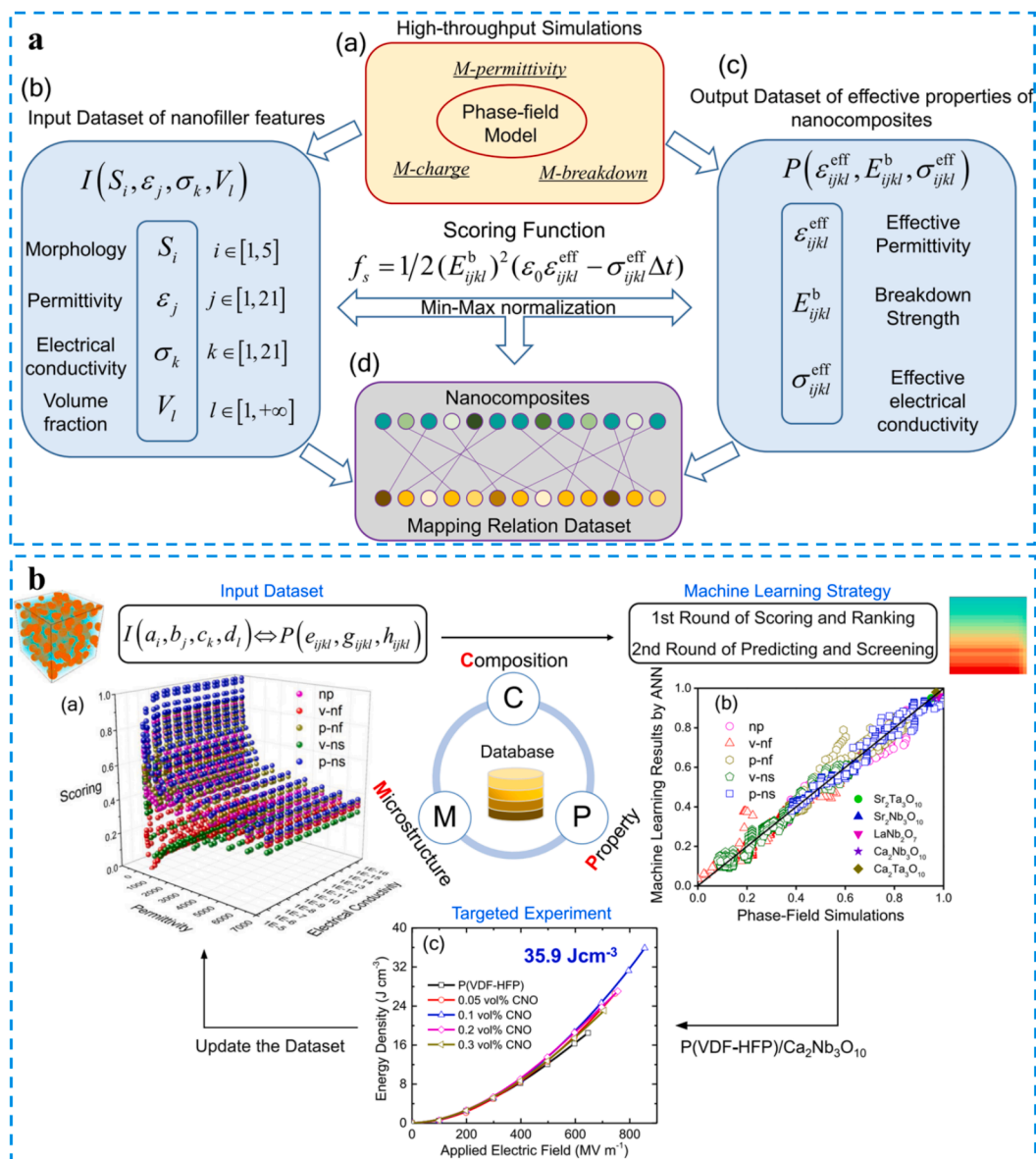


Fig. 11. High-throughput phase field simulation and ML strategy (a), and materials research paradigm (b) [149].

these issues, Shen et al. developed an integrated phase field model to simulate the dielectric response, charge transport, and breakdown processes of polymer nanocomposites [149], as displayed in Fig. 11. They formulated a ML strategy based on high-throughput computational results to evaluate energy storage capacity. Using 6615 high-throughput computational results, an input dataset containing features such as nanofiller morphology, dielectric constant, conductivity, and volume fraction, and an output dataset containing performance parameters like effective dielectric constant, breakdown strength, and effective conductivity were constructed. A ML model was built using the back-propagation neural network (BPNN) algorithm to assess the energy storage capacity of 2205 different nanocomposites. Through the ML strategy, some two-dimensional perovskite nanosheets with high energy storage potential, such as $\text{Ca}_2\text{Nb}_3\text{O}_{10}$, were identified. Experimental preparation of P(VDF-HFP)/ $\text{Ca}_2\text{Nb}_3\text{O}_{10}$ nanocomposite films validated the predictions of the ML model, exhibiting nearly double the highest discharge energy density of the original polymer, reaching 35.9 J cm^{-3} . This approach of using ML-assisted research paradigm to address the prediction and design of dielectric performance of polymer nanocomposites provides insights and methods for the characterization and development of high-energy density nanocomposites in the future.

In addition, a method combining MD simulations and ML techniques has been used to predict the mechanical properties of graphene-reinforced aluminum-based nanocomposites [150]. ANN and SVM models can capture key influencing factors of material performance, such as volume fraction of nanoparticles, orientation angles, and temperature. These models not only provide predictive results comparable to traditional MD simulations but also handle larger datasets in shorter times, offering an efficient tool for material design and optimization. Recently, a deep learning-based constitutive model has been developed to analyze the cyclic viscoelastic-viscoplastic damage behavior of water-containing nanoparticle/epoxy resin nanocomposites [151]. The LSTM can capture rate-dependent behavior of materials and integrate it into finite element analysis, significantly improving computational efficiency. Similarly, LSTM networks have been developed to characterize the wear and corrosion behavior of magnesium-cerium dioxide nanocomposites [152].

3.3. ML-assisted preparation strategies for nanomaterials

The preparation process of nanomaterials typically involves multiple complex steps and parameters that need to be optimized to achieve the desired material properties. ML can assist researchers in quickly identifying the best preparation conditions and parameters by analyzing a vast amount of experimental data and simulation results.

3.3.1. Preparation process optimization

The preparation process of nanomaterials involves the control of numerous parameters, including the ratio of raw materials, reaction temperature, reaction time, and solvent selection. Traditional methods for synthesizing nanomaterials often require trial and error to find the best preparation conditions, which necessitates a significant number of experiments and accumulation of experience. ML methods can learn from various parameters in the nanomaterial preparation process, establish an association model between parameters and structural characteristics of nanomaterials, and achieve real-time control and adjustment of the reaction process by learning from real-time monitoring data during the preparation of nanomaterials, thereby optimizing the preparation process.

Due to graphene's ultra-thin thickness and excellent mechanical properties, graphene with artificial pores shows great potential in many applications [153], such as seawater desalination. The geometric shape of nanopores in nano-porous two-dimensional materials is a major factor affecting the performance of seawater desalination [154]. Finding the optimal graphene nanopore structure remains a challenge because the discovery of the optimal nanopores usually involves a large number of

experiments or simulations, which are not only costly but also time-consuming. ML strategies can address the limitations of traditional methods. Studies have shown that deep reinforcement learning (DRL) can train an agent to find the optimal strategy that maximizes future rewards through interaction with the environment [155,156], and CNNs can extract features from the geometric characteristics of graphene nanopores and directly predict water flux and ion rejection rates [157]. An AI framework combining DRL and CNNs has been proposed [158] for discovering the most effective graphene nanopores to achieve efficient desalination of water resources, as illustrated in Fig. 12(a). The DRL agent creates nanopores through a series of decisions (whether to remove carbon atoms and which atoms to remove), while the CNN model is used to extract features from the geometric characteristics of graphene nanopores and predict their water flux and ion rejection rates. With this method, it is possible to create and evaluate 7999 different nanopores in one week, which would take about 33 years using traditional MD simulations. The research results show that the graphene nanopores created by AI achieve higher water flux while maintaining a similar ion rejection rate. The common feature of these high-performance nanopores is a semi-elliptical shape and rough edges, which help to maximize water flux while maintaining a high ion rejection rate. By combining the strengths of DRL and CNN, not only is the preparation process of nanopores precisely controlled, but the predictive accuracy and interpretability of the model are also improved, which can be applied to gas permeation and molecular translocation of two-dimensional nanomaterials.

The ML methods can also be applied to the preparation of nanocrystals in complex chemical systems [159]. Most studies treat the preparation process as a black box, directly learning from experimental parameters to reaction outcomes, but the correlations predicted are difficult to translate into physical or theoretical findings. In addition, the preparation process of most complex nanoparticles heavily relies on their kinetic processes [160]. Understanding the time-dependence of reaction progress and dynamics is crucial for understanding the progress and dynamics of the reaction. Existing solvable models find it difficult to describe the connections between different reactions involved, making the preparation of nanoparticles particularly challenging [161]. Numerical models of complex synthetic reaction networks can be parameterized based on experimental concentration time series data to determine physical parameters. ML technology provides an alternative method to effectively explore changes in concentration over time and reduce the risk of overfitting through regularization or cross-validation [162]. A ML framework has been used to predict and understand the nucleation and growth of two-dimensional perovskite nanosheets [163]. The formation process of nanomaterials at the nanoscale includes one by layer-by-layer chemical exfoliation from lead bromide nanocrystals, and the other by direct nucleation from precursor substances. The former uses phenomenological kinetic analysis to study the exfoliation process, while the latter uses ML to study direct nucleation and growth. Stopped flow preparation experiments were used to obtain time series spectral information, which provided insights into the nanosheet formation mechanism at a millisecond-level resolution. The constructed model combined the description of the optical absorption spectrum of the nanosheets with the kinetic behavior of nucleation and growth, as depicted in Fig. 12(b). By fitting the model to experimental data and evaluating the accuracy of extrapolated cross-validation predictions, it was possible to determine the physical parameters of the system and elucidate the formation mechanism of $\text{OLA}_2\text{PbBr}_4$ nanosheets in relation to the preparation method. Therefore, by combining traditional kinetic analysis and ML methods, a way has been provided to understand and predict the design and preparation of nanomaterials and to decipher complex chemical systems.

Additionally, ML has significant applications in the design and optimization of thin-film nanocomposite membranes (TFNs), especially in the process of organic solvent nanofiltration (OSN). ANN models estimate permeation flux and anti-fouling performance, with correlation

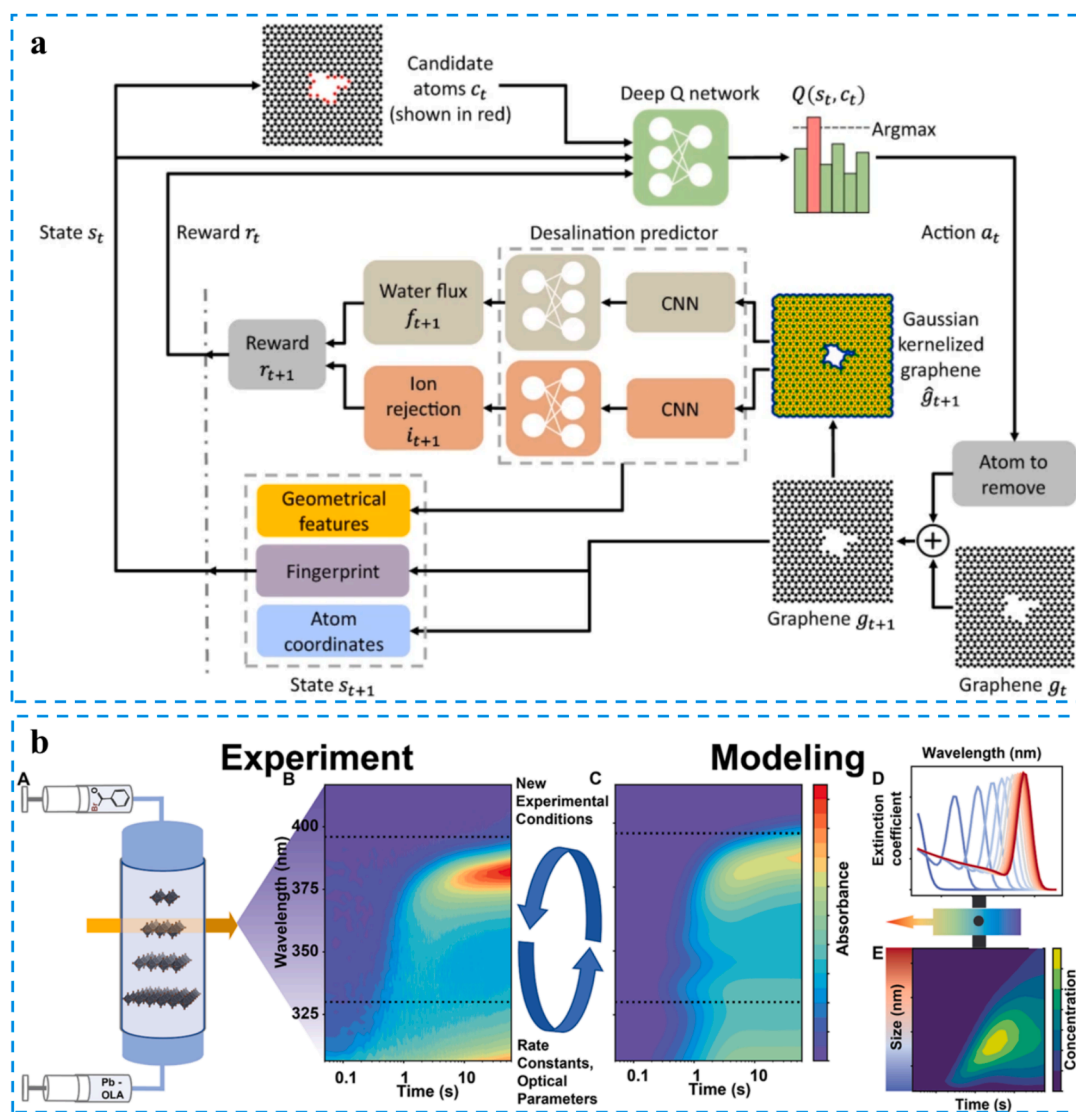


Fig. 12. ML for the optimization of nanomaterial preparation processes. (a) An AI framework composed of CNN and DRL used to find effective graphene nanopores [158], (b) Comparison of ML models with experiments [163].

coefficients all above 0.94 [52]. The bayesian regularization (BR) training algorithm has shown excellent performance in preventing overfitting. By comparing four models-linear models, SVM, boosted tree (BT), and ANNs, it was found that the BT model demonstrated the best predictive accuracy in evaluating the relative permeability (RP) and relative selectivity (RS) of TFN membranes in OSN [164]. Actively exploring the collection of more data, further optimizing ML models, and identifying more factors that affect the performance of TFN membranes can lead to a more efficient and economical membrane separation process.

3.3.2. Quality control and monitoring

Quality control and monitoring of nanomaterials are crucial in the preparation process of nanomaterials. Previous quality control methods mainly relied on laboratory testing and manual observation, which had issues such as long testing cycles, high costs, and difficulty in achieving real-time monitoring. To address these issues, ML methods are used to optimize the quality control and monitoring of nanomaterials. ML can learn from a large amount of quality monitoring data, establish quality control models, and achieve rapid and accurate monitoring and control of the quality of nanomaterials. This, in turn, enhances the quality stability of nanomaterials and promotes their application in various fields.

Sensors based on nanomaterials can respond to specific odor molecules by changing their electrical resistance or other physical properties to indicate the presence of these molecules. It is evident that machine learning algorithms can not only quickly and accurately process data obtained from sensors but also improve the fabrication of nanomaterials, making them more suitable for specific application scenarios, such as body odor detection. In many public scenarios, the detection and recognition of human body odor play a key role [165], such as in tracking suspects. Most current research on human body odor focuses on the detection of single gases [166], and there is relatively little research on the rapid detection and recognition of the overall human body odor. Traditional gas detection technologies often rely on expensive detection equipment [167], which, although mature, have shortcomings in portability and immediate detection. Based on the issues of high power consumption, slow response, and poor stability of existing sensors [168–170], decorating precious metals and metal oxide nanocrystals on MXene nanosheets to form nanocomposites can effectively improve the performance of gas sensors [171,172]. The issues of easy oxidation and poor dispersibility of MXene materials in non-polar phases have also been proven to be improved by adding organic metal halide perovskites as a second-phase material [173]. Based on this, a simple, low-cost, high-performance instant detection platform (IDP) has been developed

[174], which consists of a bionic sensor array (BSA) and a signal processing platform, capable of mimicking the human olfactory system, using multi-dimensional sensing signals and ML algorithms to identify the odor characteristics of different individuals, thus achieving human recognition. To construct a high-performance BSA, an in-situ growth strategy was used to prepare a library of MXene and perovskite nanocomposites. These materials exhibit higher performance than the original MXene due to their unique Schottky barrier structure, which is beneficial for converting the characteristics of human body odor into electrical signals. With the assistance of ML algorithms, the accuracy of volunteer breath detection and clothing odor detection of the IDP reached 69.2% and 51.1% respectively. This indicates that the IDP can effectively be used for the detection of human body odor and the recognition of individuals, providing an instant, convenient, and

accurate prototype for human recognition in more complex application scenarios. By comparing different ML algorithms, such as classification and regression tree (CART), logistic regression, KNN algorithm, SVM, LDA, and naive bayes model (NBM), it was found that the CART algorithm has the highest accuracy in identifying human breath odor. Overall, the good gas-sensing performance of the IDP is mainly attributed to improvements in material manufacturing, data collection, and processing (as indicated in Fig. 13(a)), which has given it an advantage in gas detection and identification.

Large amounts of industrial dyes are discharged into the environment through wastewater [175], which can reduce the penetration of sunlight in water bodies, severely disrupt the lives of aquatic organisms, and subsequently affect the balance of the entire ecosystem. In particular, dyes containing amines are toxic and hazardous, potentially

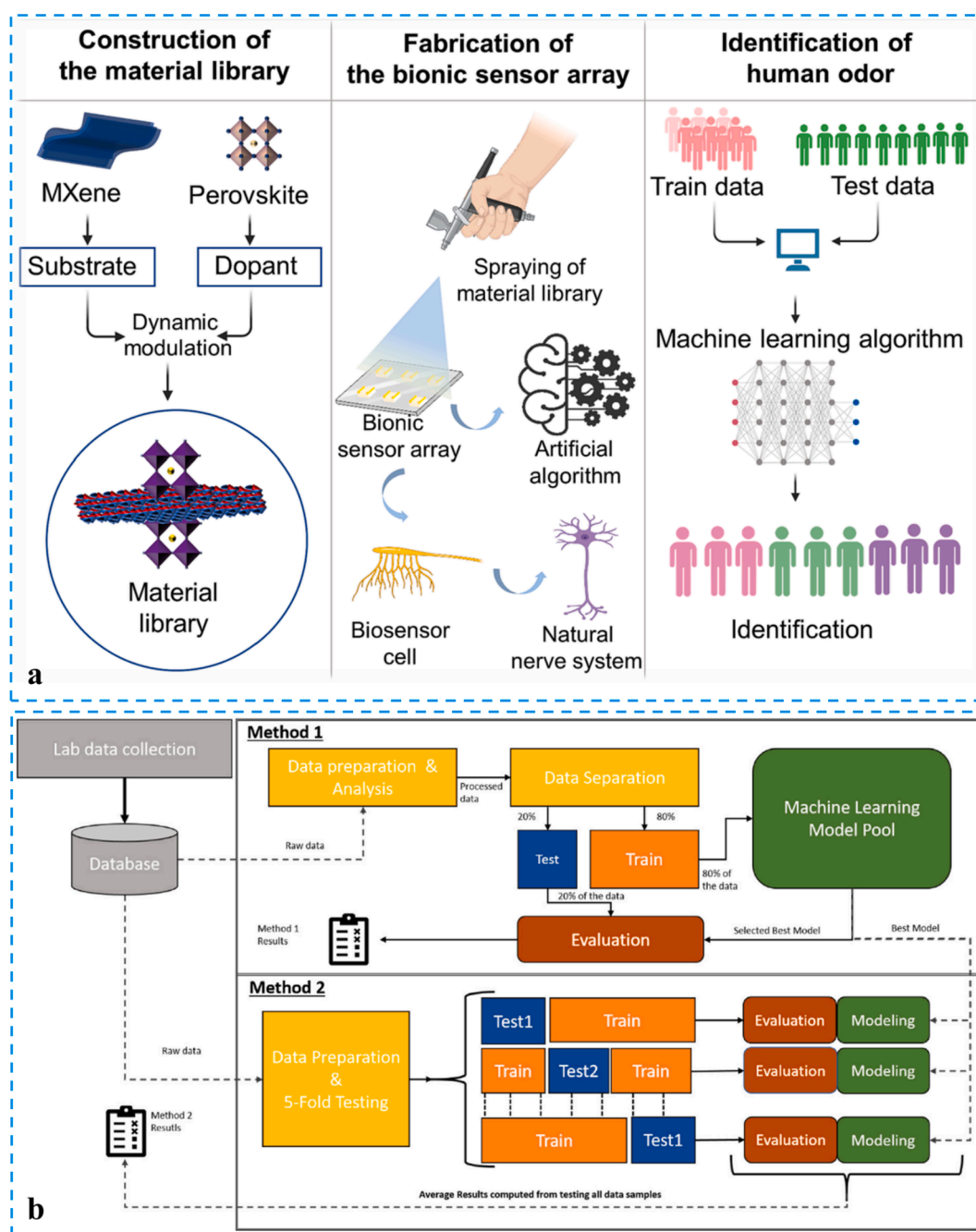


Fig. 13. ML-assisted quality monitoring of nanomaterials. (a) The process of enhancing the performance of IDP for gas detection and identification [174], (b) The workflow of a ML model [181], used for monitoring the adsorption performance of PGB-LTH nanocomposites for water pollutants.

causing a variety of health issues [176]. It is very necessary to properly treat water contaminated with dyes before reuse, and adsorption treatment is one of the most commonly used methods [177,178]. However, the efficiency of the adsorption process largely depends on the adsorbent used. Common adsorbents still have issues such as high resource consumption and low efficiency in treating dye wastewater. Utilizing ML methods to assist experimental work can save a lot of time, resources, and effort, and effectively predict adsorption performance by analyzing the complex relationships between the properties of adsorbents and the pollutants to be removed [179,180]. However, there is a lack of large datasets with detailed information about the properties of adsorbents and dyes in the research process of adsorption, and developing large datasets is time-consuming and expensive. A ML model has been developed [181] that predicts the adsorption performance of polyethyleneimine/graphene oxide/bentonite clay/layered triple hydroxide (PGB-LTH) nanocomposites for anionic and cationic dyes from aqueous solutions, considering various factors such as surface area, pore size, pH value, temperature, molecular weight, functional groups, and the charge of the adsorbent molecules, as shown in Fig. 13(b). The adsorption performance of PGB-LTH for two harmful water pollutants, namely methyl orange (MO) and crystal violet (CV), is superior to that of its original materials (i.e., graphene oxide, bentonite, and MgFeAl-LTH). The ML model was evaluated using Bagging and Boosting methods. The Extra Trees model demonstrated a correlation coefficient as high as 99% in the testing phase, along with low computational root mean squared error (RMSE) and mean absolute error (MAE), proving the model's strong capability in effectively simulating and predicting the discussed adsorption process.

ML plays an important role in improving the efficiency of nanomaterial identification in photothermal hydrogen production. Traditional nanomaterial identification methods often involve time-consuming and costly trial-and-error experiments, while ML technology can accelerate and optimize the identification process by utilizing large datasets. Advanced algorithms such as deep learning, SVMs, and RFs enable ML models to learn complex patterns and correlations within data. Combining nanotechnology, photochemistry, and principles of ML can lead to the development of efficient materials for sustainable energy production [182]. The enhanced application of ML technology not only improves the efficiency of the identification process but also helps in discovering new combinations and structures of nanomaterials with excellent photothermal properties, thereby accelerating the development of sustainable energy technologies and contributing to the global transition towards a cleaner and more efficient energy landscape. Furthermore, CNNs have been effectively used for defect detection in additive manufacturing processes, demonstrating the potential for non-destructive quality assurance in nanomaterials production [183]. Integration of ML with high-throughput experimental platforms facilitates rapid generation of synthesis data, enabling real-time tuning of synthesis parameters to achieve desired material properties [184]. Combining the optimization capabilities of bayesian optimization with regression modelling by deep neural networks provided insights into the relationship between chemical composition and spectral properties of nanoparticles. By analyzing data from high-throughput experiments, ML models can guide the preparation of nanoparticles with desired properties, thereby simplifying the materials development process [185]. The ability to predict and control nanoparticle properties demonstrates the transformative impact of ML in nanomaterials research and manufacturing. The use of ML in this context helps to achieve a higher level of control over the preparation process, leading to the production of nanomaterials with properties tailored to specific applications.

In summary, the application of ML in the preparation of nanomaterials is of significant importance. Through the analysis of two aspects, optimization of the preparation process and monitoring of quality control, we can see that ML methods can guide the preparation of nanomaterials by learning from a large amount of data, establishing models to optimize the preparation process, and achieve quality

monitoring and control. The integration of ML technology with other advanced technologies plays a key role in the preparation of nanomaterials, bringing new opportunities and challenges to the research and application of nanomaterials.

4. Future challenges and perspectives

The demand for high-performance nanomaterials across various application fields is becoming increasingly urgent. How to design and manufacture nanomaterials with outstanding performance to fully leverage their advantages has become one of the core themes of nanomaterial research. Most nanomaterial-related studies still rely on traditional empirical knowledge, which often suffers from low efficiency, instability, and long cycles, preventing nanomaterials from achieving a globally optimal solution. The emergence of ML technology has brought new ideas and methods to the development of nanomaterials. Currently, ML shows great potential in the application of nanomaterials in multiple fields. The advantages of ML in assisting nanomaterials are reflected in the following four aspects: First, by optimizing the design process of nanomaterials, it accelerates the research and development process and improves the cost-effectiveness of nanomaterials; Second, it helps researchers accurately characterize and analyze data such as spectroscopy, microscopic images, and material property testing, promoting the discovery of new nanomaterials; Third, it quickly identifies the best preparation conditions and parameters, improving the quality and yield of nanomaterials, and achieving quality control and monitoring of nanomaterials; Fourth, it overcomes the limitations of traditional methods and promotes the application of nanomaterials in various fields. Therefore, ML plays an important role in the discovery, design, characterization, preparation, analysis, optimization, simulation, prediction, and application of nanomaterials.

The combination of nanomaterials and ML technology brings great opportunities to various fields but also faces some challenges. Firstly, the design and preparation of nanomaterials is a complex process that requires consideration of various factors, such as the material's structure, performance, and application environment. ML needs to overcome difficulties in data acquisition and model training, as well as issues of model interpretability and reliability, in the process of accelerating the design and optimization of nanomaterials. Secondly, the application of nanomaterials needs to consider their performance and interactions at different scales, which poses challenges to the design and application of ML algorithms. In addition, the safety and environmental friendliness of nanomaterials are also important considerations, requiring the integration of ML technology with related fields to address these challenges. To fully leverage the advantages of ML technology in nanomaterial science, potential future development directions can be summarized as follows.

(1) Acquisition of accurate and diverse data

The performance of ML algorithms largely depends on the quality of the data. In the field of nanomaterials, due to the complexity and diversity of experimental conditions, experimental data often contain noise and bias, necessitating preprocessing and cleansing of the data. Accurately and effectively cleaning noise signals from vast datasets, extracting, and separating relevant and diverse feature data to ensure the effective operation of ML is a major challenge faced. Additionally, due to the peculiarities of nanomaterials, the associated datasets are often small, which may lead to insufficient model generalization capabilities, meaning the models perform poorly when confronted with new, unseen data. The complexity of nanomaterials requires descriptors to be accurate and comprehensive, ensuring that ML models can capture the subtle structural and performance changes of the materials. However, collecting these high-quality descriptors is a time-consuming and costly task, especially when dealing with a wide variety of nanomaterials, further increasing the difficulty and cost of research. Therefore, a major challenge in applying ML to nanomaterials is managing small datasets and collecting descriptors. Strengthening the collection and

organization of data related to nanomaterials, effectively capturing the inherent characteristics and behaviors of nanomaterials, establishing robust nanomaterial databases, and developing corresponding intelligent algorithms and data processing techniques can provide more data support for nanomaterial applications. Additionally, clarifying the impact of various factors on nanomaterials and establishing relationship patterns between data and the structure and performance of nanomaterials may require leveraging the advantages of deep learning in feature data extraction to achieve precise design and performance prediction of nanomaterials under complex conditions.

(2) Development of interpretable ML algorithms and models

ML models are typically founded on statistical data analysis, and their predictive outcomes can often be challenging to interpret. In the design and characterization of nanomaterials, it is crucial to select the appropriate ML algorithm based on specific application scenarios and data characteristics to enhance interpretability. ML algorithms have become tools for addressing specific challenges in nanomaterial research, with their effectiveness varying according to the nature of the problem. Deep learning and neural networks excel at handling large-scale data and have been utilized for rapid data analysis in combinatorial nanoparticle syntheses. They are valuable for predicting the performance of nanomaterials because they can process low-cost, low-precision data from extensive datasets. Reinforcement learning is renowned for its precision in reverse design, offering a pathway for the “performance prediction-material design” model of nanomaterials. Bayesian optimization is effective for driving high-throughput experimental loops, particularly when initiated with sparse datasets. SVMs can handle classification and regression tasks in nanomaterial research and are effective when the relationship between input and output is not linear. Generative models are explored for inverse design problems in materials science. In conclusion, the selection of ML algorithms in nanomaterial research should follow the specific requirements of the task, including the size and complexity of the dataset, the need for interpretability, and the desired outcome, whether it be prediction, classification, or inverse design. Furthermore, the development of new ML algorithms and models tailored to the unique needs of the nanomaterials field can improve the accuracy and generalization capabilities of the models. To promote the modeling and prediction accuracy of nanomaterial design and performance, it is essential to strengthen the interpretation and verification of ML models and to develop ML algorithms with strong interpretability and high reliability suitable for nanomaterials, thereby enhancing the application effect and credibility of ML in the field. The development of new ML algorithms and models with strong interpretability will likely become a new trend in the development of high-performance nanomaterials. For instance, multi-precision deep neural networks can capture low-cost, low-precision data from a large amount of complex data, showing potential application advantages in the performance prediction of nanomaterials.

(3) Establishment of user-friendly and high-performance tools and platforms

ML technology can obtain more accurate and reliable results in a shorter time, but its use is not particularly widespread, mainly because not all researchers have the knowledge or skills needed to fully utilize ML technology, which may hinder the integration of these advanced optimization algorithms into daily research applications and practical life. To facilitate the use of ML technology by researchers with different levels of experience and professional knowledge at any time, establishing user-friendly tools and platforms based on ML technology, tailored to the performance characteristics and design optimization strategies of nanomaterials, is an effective way to quickly promote the development of nanomaterial science. ML can reduce errors and improve overall accuracy and efficiency in the field of nanomaterials. By integrating ML with intelligent media applications, it is conducive to the development of smart devices. Pattern recognition and detection based on ML technology can develop wearable sensors based on nanomaterials that can monitor health status in real-time. For example, ML-assisted sensing

platforms can also be used for on-site environmental monitoring to help detect various food and environmental analytes, developing smart devices to assess compounds that threaten health. In addition, ML technology can also be used to establish a data sharing platform in the field of nanomaterials, promoting the standardization and specification of data, improving the usability and quality of data.

(4) Construction of a reinforcement system of multidisciplinary and multi-technological integration

The application of ML in the field of nanomaterials involves the interdisciplinary integration of materials science, computer science, environmental science, medical science, energy science, robotics, and other disciplines. It is essential to fully understand the inherent properties of nanomaterials and strengthen the interdisciplinary cooperation and exchange between ML and related sciences of nanomaterials. This promotes knowledge sharing and technological innovation between different fields, bringing more opportunities and challenges to the application and development of nanomaterials. With the advent of the new generation of information technology, big data, and the era of the Internet of Things, various new technologies are rapidly developing in the synthesis and processing of nanomaterials, quality control, and performance optimization. The combination of these technologies with ML technology can quickly establish data-driven models, comprehensively consider various uncertainty factors, and achieve the purpose of all-round prediction and diagnosis. For example, the combination of digital twin technology and ML technology can achieve the complementarity of digital models and actual states, improve nanomaterial structural design strategies, and predict and evaluate their performance. In addition, the intelligent algorithms of ML and multi-agent technology can empower the manufacturing of high-performance nanomaterials with complex structures using printing technology. We have reason to believe that in the future, ML technology will serve as a bridge to closely connect various disciplines and technologies in the field of nanomaterials, injecting new vitality into the development of nanomaterials.

In summary, although research on ML-assisted nanomaterials is not yet mature, it has shown unimaginable potential. The application of ML in the field of nanomaterials provides a more rapid, more effective, and more intelligent method of data analysis and modeling for nanomaterial research, improving the accuracy of material design, characterization, and performance prediction, and promoting the rapid development of nanomaterial research. Future research can start from aspects such as data acquisition and processing, algorithm and model optimization, tool and platform development, and the establishment of a reinforcement system to promote the multi-faceted application of ML in the field of nanomaterials and bring new opportunities and breakthroughs to the development of various fields. It is hoped that this overview can provide some reference and inspiration for researchers in related fields.

CRediT authorship contribution statement

Liang Yang: Writing – review & editing, Writing – original draft, Visualization, Methodology, Data curation. **Hong Wang:** Writing – review & editing, Writing – original draft, Investigation, Formal analysis. **Deying Leng:** Visualization, Investigation, Formal analysis. **Shipeng Fang:** Visualization, Formal analysis. **Yanning Yang:** Investigation, Formal analysis. **Yurun Du:** Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

No data was used for the research described in the article.

References

- [1] J. Zhou, A.I. Chizhik, S. Chu, D. Jin, Single-particle spectroscopy for functional nanomaterials, *Nature* 579 (7797) (2020) 41–50, <https://doi.org/10.1038/s41586-020-2048-8>.
- [2] J. Liu, R. Li, B. Yang, Carbon Dots: A New Type of Carbon-Based Nanomaterial with Wide Applications, *ACS Central Science* 6 (12) (2020) 2179–2195, <https://doi.org/10.1021/acscentsci.0c01306>.
- [3] K. Liu, H. Du, T. Zheng, W. Liu, M. Zhang, H. Liu, X. Zhang, C. Si, Lignin-containing cellulose nanomaterials: preparation and applications, *Green Chemistry* 23 (24) (2021) 9723–9746, <https://doi.org/10.1039/D1GC02841C>.
- [4] D. Han, C. Li, C. Jiang, X. Jin, X. Wang, R. Chen, J. Cheng, P. Duan, Endowing inorganic nanomaterials with circularly polarized luminescence, *Aggregate* 3 (1) (2022) e148, <https://doi.org/10.1002/agt2.148>.
- [5] X. Chen, Y. Liu, B. Wang, X. Liu, C. Lu, Understanding role of microstructures of nanomaterials in electrochemiluminescence properties and their applications, *TrAC Trends in Analytical Chemistry* 162 (2023) 117030, <https://doi.org/10.1016/j.trac.2023.117030>.
- [6] Z. Pei, H. Lei, L. Cheng, Bioactive inorganic nanomaterials for cancer theranostics, *Chemical Society Reviews* 52 (6) (2023) 2031–2081, <https://doi.org/10.1039/D2CS00352J>.
- [7] L. Yang, H. Wang, F. Zhang, Y. Yang, S. Qu, D. Leng, X. Ma, Synthetic technologies, property enhancements and versatile applications of calcium copper titanate: A review, *Nano Energy* 114 (2023) 108609, <https://doi.org/10.1016/j.nanoen.2023.108609>.
- [8] M. Li, S. Kuang, Y. Kang, H. Ma, J. Dong, Z. Guo, Recent advances in application of iron-manganese oxide nanomaterials for removal of heavy metals in the aquatic environment, *Science of the Total Environment* 819 (2022) 153157, <https://doi.org/10.1016/j.scitotenv.2022.153157>.
- [9] Q. Wei, Y. Pan, Z. Zhang, S. Yan, Z. Li, Copper-based nanomaterials for biomedical applications, *Chemical Engineering Journal* 483 (2024) 149040, <https://doi.org/10.1016/j.cej.2024.149040>.
- [10] N. Wang, R. Zhang, K. Liu, Y. Zhang, X. Shi, W. Sand, B. Hou, Application of nanomaterials in antifouling: A review, *Nano, Materials Science* (2024), <https://doi.org/10.1016/j.nanoms.2024.01.009>.
- [11] L. Li, T. Wang, Y. Zhong, R. Li, W. Deng, X. Xiao, Y. Xu, J. Zhang, X. Hu, Y. Wang, A review of nanomaterials for biosensing applications, *Journal of Materials Chemistry B* 12 (5) (2024) 1168–1193, <https://doi.org/10.1039/D3TB02648E>.
- [12] P. Raccuglia, K.C. Elbert, P.D.F. Adler, C. Falk, M.B. Wenny, A. Mollo, M. Zeller, S.A. Friedler, J. Schrier, A.J. Norquist, Machine-learning-assisted materials discovery using failed experiments, *Nature* 533 (7601) (2016) 73–76, <https://doi.org/10.1038/nature17439>.
- [13] Y. Wei, J. Wu, Y. Wu, H. Liu, F. Meng, Q. Liu, A.C. Midgley, X. Zhang, T. Qi, H. Kang, R. Chen, D. Kong, J. Zhuang, X. Yan, X. Huang, Prediction and Design of Nanozymes using Explainable Machine Learning, *Advanced Materials* 34 (27) (2022) 2201736, <https://doi.org/10.1002/adma.202201736>.
- [14] A. Rahman, P. Deshpande, M.S. Radue, G.M. Odegard, S. Gowtham, S. Ghosh, A. D. Spear, A machine learning framework for predicting the shear strength of carbon nanotube-polymer interfaces based on molecular dynamics simulation data, *Composites Science and Technology* 207 (2021) 108627, <https://doi.org/10.1016/j.compscitech.2020.108627>.
- [15] K.T. Butler, D.W. Davies, H. Cartwright, O. Isayev, A. Walsh, Machine learning for molecular and materials science, *Nature* 559 (7715) (2018) 547–555, <https://doi.org/10.1038/s41586-018-0337-2>.
- [16] T. Li, C. Zhang, X. Li, Machine learning for flow batteries: opportunities and challenges, *Chemical Science* 13 (17) (2022) 4740–4752, <https://doi.org/10.1039/D2SC00291D>.
- [17] J.L. Raymond, J.F. Medina, Computational Principles of Supervised Learning in the Cerebellum, *Annual Review of Neuroscience* 41 (Volume 41, 2018) (2018) 233–253, <https://doi.org/https://doi.org/10.1146/annurev-neuro-080317-061948>.
- [18] S. Milani, N. Topin, M. Veloso, F. Fang, Explainable Reinforcement Learning: A Survey and Comparative Review, *ACM Computing Surveys* 56(7 %J ACM Comput. Surv.) (2024) Article 168, <https://doi.org/10.1145/3616864>.
- [19] J.E. Gerken, J. Aronsson, O. Carlsson, H. Linander, F. Ohlsson, C. Petersson, D. Persson, Geometric deep learning and equivariant neural networks, *Artificial Intelligence Review* 56 (12) (2023) 14605–14662, <https://doi.org/10.1007/s10462-023-10502-7>.
- [20] C. Zhao, D. Wu, J. Huang, Y. Yuan, H.T. Zhang, R. Peng, Z. Shi, BoostTree and BoostForest for Ensemble Learning, *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45 (7) (2023) 8110–8126, <https://doi.org/10.1109/TPAMI.2022.3227370>.
- [21] M.R. Carbone, M. Topsakal, D. Lu, S. Yoo, Machine-Learning X-Ray Absorption Spectra to Quantitative Accuracy, *Physical Review Letters* 124 (15) (2020) 156401, <https://doi.org/10.1103/PhysRevLett.124.156401>.
- [22] T.T. Wong, P.Y. Yeh, Reliable Accuracy Estimates from k-Fold Cross Validation, *IEEE Transactions on Knowledge and Data Engineering* 32 (8) (2020) 1586–1594, <https://doi.org/10.1109/TKDE.2019.2912815>.
- [23] C. Zhang, C. Lu, L. Pei, J. Li, R. Wang, K. Tieu, The negative Poisson's ratio and strengthening mechanism of nanolayered graphene/Cu composites, *Carbon* 143 (2019) 125–137, <https://doi.org/10.1016/j.carbon.2018.10.097>.
- [24] S. Zhang, P. Huang, F. Wang, Graphene-boundary strengthening mechanism in Cu/graphene nanocomposites: A molecular dynamics simulation, *Materials & Design* 190 (2020) 108555, <https://doi.org/10.1016/j.matdes.2020.108555>.
- [25] Y.G. Chung, E. Haldoupis, B.J. Bucior, M. Haranczyk, S. Lee, H. Zhang, K. D. Vogiatzis, M. Milisavljevic, S. Ling, J.S. Camp, B. Slater, J.I. Siepmann, D. S. Sholl, R.Q. Snurr, Advances, Updates, and Analytics for the Computation-Ready, Experimental Metal-Organic Framework Database: CoRE MOF 2019, *Journal of Chemical & Engineering Data* 64 (12) (2019) 5985–5998, <https://doi.org/10.1021/acs.jced.9b00835>.
- [26] H. Wang, L. Yang, Y. Yang, D. Zhang, A. Tian, Highly flexible, large-deformation ionic polymer metal composites for artificial muscles: Fabrication, properties, applications, and prospects, *Chemical Engineering Journal* 469 (2023) 143976, <https://doi.org/10.1016/j.cej.2023.143976>.
- [27] S. Chong, S. Lee, B. Kim, J. Kim, Applications of machine learning in metal-organic frameworks, *Coordination Chemistry Reviews* 423 (2020) 213487, <https://doi.org/10.1016/j.ccr.2020.213487>.
- [28] K.M. Jablonka, D. Ongari, S.M. Moosavi, B. Smit, Big-Data Science in Porous Materials: Materials Genomics and Machine Learning, *Chemical Reviews* 120 (16) (2020) 8066–8129, <https://doi.org/10.1021/acs.chemrev.0c00004>.
- [29] G.S. Fanourgakis, K. Gkagkas, E. Tylianakis, E. Klontzas, G. Froudakis, A Robust Machine Learning Algorithm for the Prediction of Methane Adsorption in Nanoporous Materials, *The Journal of Physical Chemistry A* 123 (28) (2019) 6080–6087, <https://doi.org/10.1021/acs.jpca.9b03290>.
- [30] G.S. Fanourgakis, K. Gkagkas, E. Tylianakis, G. Froudakis, A Generic Machine Learning Algorithm for the Prediction of Gas Adsorption in Nanoporous Materials, *The Journal of Physical Chemistry C* 124 (13) (2020) 7117–7126, <https://doi.org/10.1021/acs.jpcc.9b10766>.
- [31] R. Anderson, J. Rodgers, E. Argueta, A. Biong, D.A. Gómez-Gualdrón, Role of Pore Chemistry and Topology in the CO₂ Capture Capabilities of MOFs: From Molecular Simulation to Machine Learning, *Chemistry of Materials* 30 (18) (2018) 6325–6337, <https://doi.org/10.1021/acs.chemmater.8b02257>.
- [32] A. Datar, Y.G. Chung, L.-C. Lin, Beyond the BET Analysis: The Surface Area Prediction of Nanoporous Materials Using a Machine Learning Method, *The Journal of Physical Chemistry Letters* 11 (14) (2020) 5412–5417, <https://doi.org/10.1021/acs.jpclett.0c01518>.
- [33] T. Zhou, Z. Song, K. Sundmacher, Big Data Creates New Opportunities for Materials Research: A Review on Methods and Applications of Machine Learning for Materials Design, *Engineering* 5 (6) (2019) 1017–1026, <https://doi.org/10.1016/j.eng.2019.02.011>.
- [34] E.H. Cho, L.-C. Lin, Nanoporous Material Recognition via 3D Convolutional Neural Networks: Prediction of Adsorption Properties, *The Journal of Physical Chemistry Letters* 12 (9) (2021) 2279–2285, <https://doi.org/10.1021/acs.jpclett.1c00293>.
- [35] N. Serov, V. Vinogradov, Artificial intelligence to bring nanomedicine to life, *Advanced Drug Delivery Reviews* 184 (2022) 114194, <https://doi.org/10.1016/j.addr.2022.114194>.
- [36] A.V. Singh, R.-S. Maharjan, A. Kanase, K. Siewert, D. Rosenkranz, R. Singh, P. Laux, A. Luch, Machine-Learning-Based Approach to Decode the Influence of Nanomaterial Properties on Their Interaction with Cells, *ACS Applied Materials & Interfaces* 13 (1) (2021) 1943–1955, <https://doi.org/10.1021/acsami.0c18470>.
- [37] N. El Yamani, E. Mariussen, M. Gromelski, E. Wyrzykowska, D. Grabarek, T. Puzyn, S. Tanasescu, M. Dusinska, E. Rundén-Pran, Hazard identification of nanomaterials: In silico unraveling of descriptors for cytotoxicity and genotoxicity, *Nano Today* 46 (2022) 101581, <https://doi.org/10.1016/j.nantod.2022.101581>.
- [38] N. Serov, V. Vinogradov, Inverse Material Search and Synthesis Verification by Hand Drawings via Transfer Learning and Contour Detection, *Small Methods* 6 (5) (2022) 2101619, <https://doi.org/10.1002/smt.202101619>.
- [39] X. Yan, J. Zhang, D.P. Russo, H. Zhu, B. Yan, Prediction of Nano-Bio Interactions through Convolutional Neural Network Analysis of Nanostructure Images, *ACS Sustainable Chemistry & Engineering* 8 (51) (2020) 19096–19104, <https://doi.org/10.1021/acssuschemeng.0c07453>.
- [40] G. Gul, R. Yildirim, N. Ileri-Ercan, Cytotoxicity analysis of nanoparticles by association rule mining, *Environmental Science: Nano* 8 (4) (2021) 937–949, <https://doi.org/10.1039/D0EN01240H>.
- [41] N. Shirokii, Y. Din, I. Petrov, Y. Seregin, S. Sirotenko, J. Razlivina, N. Serov, V. Vinogradov, Quantitative Prediction of Inorganic Nanomaterial Cellular Toxicity via Machine Learning, *Small* 19 (19) (2023) 2207106, <https://doi.org/10.1002/smll.202207106>.
- [42] S.F. Hansen, O.F.H. Hansen, M.B. Nielsen, Advances and challenges towards consumerization of nanomaterials, *Nature Nanotechnology* 15 (12) (2020) 964–965, <https://doi.org/10.1038/s41565-020-00819-7>.
- [43] F. Yu, C. Wei, P. Deng, T. Peng, X. Hu, Deep exploration of random forest model boosts the interpretability of machine learning studies of complicated immune responses and lung burden of nanoparticles, *Science, Advances* 7 (22) (2021) eabf4130.

- [44] C. Gousiadou, R.L. Marchese Robinson, M. Kotzabasaki, P. Doganis, T.A. Wilkins, X. Jia, H. Sarimveis, S.L. Harper, Machine learning predictions of concentration-specific aggregate hazard scores of inorganic nanomaterials in embryonic zebrafish, *Nanotoxicology* 15(4) (2021) 446–476. <https://doi.org/10.1080/17435390.2021.1872113>.
- [45] K. Pikula, A. Zakharenko, V. Chaika, K. Kirichenko, A. Tsatsakis, K. Golokhvast, Risk assessments in nanotoxicology: bioinformatics and computational approaches, *Current Opinion in Toxicology* 19 (2020) 1–6. <https://doi.org/10.1016/j.cotox.2019.08.006>.
- [46] J.-S. Choi, M.K. Ha, T.X. Trinh, T.H. Yoon, H.-G. Byun, Towards a generalized toxicity prediction model for oxide nanomaterials using integrated data from different sources, *Scientific Reports* 8 (1) (2018) 6110. <https://doi.org/10.1038/s41598-018-24483-z>.
- [47] Y. Pan, T. Li, J. Cheng, D. Telesca, J.L. Zink, J. Jiang, Nano-QSAR modeling for predicting the cytotoxicity of metal oxide nanoparticles using novel descriptors, *RSC Advances* 6 (31) (2016) 25766–25775. <https://doi.org/10.1039/C6RA01298A>.
- [48] N. Ye, Z. Wang, S. Wang, W.J.G.M. Peijnenburg, Toxicity of mixtures of zinc oxide and graphene oxide nanoparticles to aquatic organisms of different trophic level: particles outperform dissolved ions, *Nanotoxicology* 12 (5) (2018) 423–438. <https://doi.org/10.1080/17435390.2018.1458342>.
- [49] Y. Zhou, Y. Wang, W. Peijnenburg, M.G. Vijver, S. Balraadsing, W. Fan, Using Machine Learning to Predict Adverse Effects of Metallic Nanomaterials to Various Aquatic Organisms, *Environmental Science & Technology* 57 (46) (2023) 17786–17795. <https://doi.org/10.1021/acs.est.2c07039>.
- [50] A. Panagopoulos, K.-J. Haralambous, M. Loizidou, Desalination brine disposal methods and treatment technologies - A review, *Science of the Total Environment* 693 (2019) 133545. <https://doi.org/10.1016/j.scitotenv.2019.07.351>.
- [51] M. Qasim, M. Badrelzaman, N.N. Darwish, N.A. Darwish, N. Hilal, Reverse osmosis desalination: A state-of-the-art review, *Desalination* 459 (2019) 59–104. <https://doi.org/10.1016/j.desal.2019.02.008>.
- [52] M. Fetanat, M. Keshtiar, R. Keyikoglu, A. Khataee, R. Daiyan, A. Razmjou, Machine learning for design of thin-film nanocomposite membranes, *Separation and Purification Technology* 270 (2021) 118383. <https://doi.org/10.1016/j.seppur.2021.118383>.
- [53] R. Wang, S. Lin, Pore model for nanofiltration: History, theoretical framework, key predictions, limitations, and prospects, *Journal of Membrane Science* 620 (2021) 118809. <https://doi.org/10.1016/j.memsci.2020.118809>.
- [54] Q. Zhao, D.L. Zhao, T.-S. Chung, Nanoclays-Incorporated Thin-Film Nanocomposite Membranes for Reverse Osmosis Desalination, *Advanced Materials Interfaces* 7 (8) (2020) 1902108. <https://doi.org/10.1002/admi.201902108>.
- [55] C.S.H. Yeo, Q. Xie, X. Wang, S. Zhang, Understanding and optimization of thin film nanocomposite membranes for reverse osmosis with machine learning, *Journal of Membrane Science* 606 (2020) 118135. <https://doi.org/10.1016/j.memsci.2020.118135>.
- [56] C. Janiesch, P. Zschech, K. Heinrich, Machine learning and deep learning, *Electronic Markets* 31 (3) (2021) 685–695. <https://doi.org/10.1007/s12525-021-00475-2>.
- [57] T. Bonny, M. Kashkash, F. Ahmed, An efficient deep reinforcement machine learning-based control reverse osmosis system for water desalination, *Desalination* 522 (2022) 115443. <https://doi.org/10.1016/j.desal.2021.115443>.
- [58] H. Li, B. Zeng, T. Qiu, W. Huang, Y. Wang, G.-P. Sheng, Y. Wang, Deep learning models for assisted decision-making in performance optimization of thin film nanocomposite membranes, *Journal of Membrane Science* 687 (2023) 122093. <https://doi.org/10.1016/j.memsci.2023.122093>.
- [59] J.W. Barnett, C.R. Bilchak, Y. Wang, B.C. Benicewicz, L.A. Murdock, T. Bereau, S. K. Kumar, Designing exceptional gas-separation polymer membranes using machine learning, *Science, Advances* 6 (20) (2020) eaaz4301.
- [60] H. Li, B. Zeng, J. Tuo, Y. Wang, G.-P. Sheng, Y. Wang, Development of an improved deep network model as a general technique for thin film nanocomposite reverse osmosis membrane simulation, *Journal of Membrane Science* 692 (2024) 122320. <https://doi.org/10.1016/j.memsci.2023.122320>.
- [61] I. Pastoriza-Santos, C. Kinnear, J. Pérez-Juste, P. Mulvaney, L.M. Liz-Marzán, Plasmonic polymer nanocomposites, *Nature Reviews Materials* 3 (10) (2018) 375–391. <https://doi.org/10.1038/s41578-018-0050-7>.
- [62] J. Huang, Y. Qian, K. Evans, T. Xu, Diffusion-Dependent Nanoparticle Assembly in Thin Films of Supramolecular Nanocomposites: Effects of Particle Size and Supramolecular Morphology, *Macromolecules* 52 (15) (2019) 5801–5810. <https://doi.org/10.1021/acs.macromol.9b00362>.
- [63] E. Vargo, K.M. Evans, Q. Wang, A. Sattler, Y. Qian, J. Yao, T. Xu, Orbital Angular Momentum from Self-Assembled Concentric Nanoparticle Rings, *Advanced Materials* 33 (40) (2021) 2103563. <https://doi.org/10.1002/adma.202103563>.
- [64] K. Niendorf, B. Raeymaekers, Using supervised machine learning methods to predict microfiber alignment and electrical conductivity of polymer matrix composite materials fabricated with ultrasound directed self-assembly and stereolithography, *Computational Materials Science* 206 (2022) 111233. <https://doi.org/10.1016/j.commatsci.2022.111233>.
- [65] D.T. Ahneman, J.G. Estrada, S. Lin, S.D. Dreher, A.G. Doyle, Predicting reaction performance in C-N cross-coupling using machine learning, *Science* 360 (6385) (2018) 186–190.
- [66] E. Vargo, J.C. Dahl, K.M. Evans, T. Khan, P. Alivisatos, T. Xu, Using Machine Learning to Predict and Understand Complex Self-Assembly Behaviors of a Multicomponent Nanocomposite, *Advanced Materials* 34 (32) (2022) 2203168. <https://doi.org/10.1002/adma.202203168>.
- [67] C. Qiu, B. Wang, N. Zhang, S. Zhang, J. Liu, D. Walker, Y. Wang, H. Tian, T. R. Shrout, Z. Xu, L.-Q. Chen, F. Li, Transparent ferroelectric crystals with ultrahigh piezoelectricity, *Nature* 577 (7790) (2020) 350–354. <https://doi.org/10.1038/s41586-019-1891-y>.
- [68] Z. Liu, L. Wang, X. Yu, J. Zhang, R. Yang, X. Zhang, Y. Ji, M. Wu, L. Deng, L. Li, Z. L. Wang, Piezoelectric-Effect-Enhanced Full-Spectrum Photoelectrocatalysis in p-n Heterojunction, *Advanced Functional Materials* 29 (41) (2019) 1807279. <https://doi.org/10.1002/adfm.201807279>.
- [69] K. Shi, B. Chai, H. Zou, P. Shen, B. Sun, P. Jiang, Z. Shi, X. Huang, Interface induced performance enhancement in flexible BaTiO₃/PVDF-TrFE based piezoelectric nanogenerators, *Nano Energy* 80 (2021) 105515. <https://doi.org/10.1016/j.nanoen.2020.105515>.
- [70] Z.-H. Shen, J.-J. Wang, J.-Y. Jiang, S.X. Huang, Y.-H. Lin, C.-W. Nan, L.-Q. Chen, Y. Shen, Phase-field modeling and machine learning of electric-thermal-mechanical breakdown of polymer-based dielectrics, *Nature Communications* 10 (1) (2019) 1843. <https://doi.org/10.1038/s41467-019-09874-8>.
- [71] W. Li, T. Yang, C. Liu, Y. Huang, C. Chen, H. Pan, G. Xie, H. Tai, Y. Jiang, Y. Wu, Z. Kang, L.-Q. Chen, Y. Su, Z. Hong, Optimizing Piezoelectric Nanocomposites by High-Throughput Phase-Field Simulation and Machine Learning, *Advanced Science* 9 (13) (2022) 2105550. <https://doi.org/10.1002/advsc.202105550>.
- [72] H. Jahandideh, J.-R. Macairan, A. Bahmani, M. Lapointe, N. Tufenkji, Fabrication of graphene-based porous materials: traditional and emerging approaches, *Chemical Science* 13 (31) (2022) 8924–8941. <https://doi.org/10.1039/d2sc01786e>.
- [73] L. Yang, H. Wang, High-performance electrically responsive artificial muscle materials for soft robot actuation, *Acta Biomaterialia* (2024). <https://doi.org/10.1016/j.actbio.2024.07.016>.
- [74] A. Bahmani, P.A. Comeau, J. Montesano, T.L. Willett, Extrudable hydroxyapatite/plant oil-based biopolymer nanocomposites for biomedical applications: Mechanical testing and modeling, *Materials & Design* 174 (2019) 107790. <https://doi.org/10.1016/j.matdes.2019.107790>.
- [75] S.M. Mirkhalaf, E.H. Eggels, T.J.H. van Beurden, F. Larsson, M. Fagerström, A finite element based orientation averaging method for predicting elastic properties of short fiber reinforced composites, *Composites Part B: Engineering* 202 (2020) 108388. <https://doi.org/10.1016/j.compositesb.2020.108388>.
- [76] A. Shafiei, J.W. Pro, R. Martini, F. Barthelat, The very hard and the very soft: Modeling bio-inspired scaled skins using the discrete element method, *Journal of the Mechanics and Physics of Solids* 146 (2021) 104176. <https://doi.org/10.1016/j.jmps.2020.104176>.
- [77] J. Weng, W. Wen, H. Cui, B. Chen, Micromechanical analysis of composites with fibers distributed randomly over the transverse cross-section, *Acta Astronautica* 147 (2018) 133–140. <https://doi.org/10.1016/j.actaastro.2018.03.056>.
- [78] G. Li, F. Sharifpour, A. Bahmani, J. Montesano, A new approach to rapidly generate random periodic representative volume elements for microstructural assessment of high volume fraction composites, *Materials & Design* 150 (2018) 124–138. <https://doi.org/10.1016/j.matdes.2018.04.031>.
- [79] W. Tian, X. Chao, M.W. Fu, L. Qi, An advanced method for efficiently generating composite RVEs with specified particle orientation, *Composites Science and Technology* 205 (2021) 108647. <https://doi.org/10.1016/j.compscitech.2021.108647>.
- [80] A. Bahmani, R.Y. Nooraie, T.L. Willett, J. Montesano, A sequential mobile packing algorithm for micromechanical assessment of heterogeneous materials, *Composites Science and Technology* 237 (2023) 110008. <https://doi.org/10.1016/j.compscitech.2023.110008>.
- [81] S. Szymkuć, E.P. Gajewska, T. Klucznik, K. Molga, P. Dittwald, M. Startek, M. Bajczyk, B.A. Grzybowski, Computer-Assisted Synthetic Planning: The End of the Beginning, *Angewandte Chemie International Edition* 55 (20) (2016) 5904–5937. <https://doi.org/10.1002/anie.201506101>.
- [82] O.A. von Lilienfeld, K. Burke, Retrospective on a decade of machine learning for chemical discovery, *Nature Communications* 11 (1) (2020) 4895. <https://doi.org/10.1038/s41467-020-18556-9>.
- [83] Q. Wu, C. Liu, C. Cui, L. Li, L. Yang, Y. Liu, H. Safari Yazd, S. Xu, X. Li, Z. Chen, W. Tan, Plasmon Coupling in DNA-Assembled Silver Nanoclusters, *Journal of the American Chemical Society* 143(36) (2021) 14573–14580. <https://doi.org/10.1021/jacs.1c04949>.
- [84] A. González-Rosell, C. Cerretani, P. Mastracco, T. Vosch, S.M. Copp, Structure and luminescence of DNA-templated silver clusters, *Nanoscale Advances* 3 (5) (2021) 1230–1260. <https://doi.org/10.1039/D0NA01005G>.
- [85] F. Zhai, Y. Guan, B. Zhu, S. Chen, R. He, Intraparticle and Interparticle Transferable DNA Walker Supported by DNA Micelles for Rapid Detection of MicroRNA, *Analytical Chemistry* 93 (36) (2021) 12346–12352. <https://doi.org/10.1021/acs.analchem.1c02104>.
- [86] F. Xu, T. Qing, Z. Qing, DNA-coded metal nano-fluorophores: Preparation, properties and applications in biosensing and bioimaging, *Nano Today* 36 (2021) 101021. <https://doi.org/10.1016/j.nantod.2020.101021>.
- [87] L. Qin, K. Zhang, B. Feng, P. Zhang, T. Qing, J. Fei, Proximity sequence-dependent spectral conversion of silver nanoclusters and construction of ratiometric nanoprobes, *Chemical Engineering Journal* 441 (2022) 136001. <https://doi.org/10.1016/j.cej.2022.136001>.
- [88] R. Nagda, S. Park, I.L. Jung, K. Nam, H.C. Yadavalli, Y.M. Kim, K. Yang, J. Kang, P.W. Thulstrup, M.J. Bjerrum, M. Cho, T.-H. Kim, Y.H. Roh, P. Shah, S.W. Yang, Silver Nanoclusters Serve as Fluorescent Rivets Linking Hoogsteen Triplex DNA and Hairpin-Loop DNA Structures, *ACS Nano* 16 (8) (2022) 13211–13222. <https://doi.org/10.1021/acsnano.2c06631>.
- [89] C. Li, Z. Chen, Y. Zhang, J. He, R. Yuan, W. Xu, Guanine-Lighting-Up Fluorescence Biosensing of Silver Nanoclusters Populated in Functional DNA Constructs by a

- pH-Triggered Switch, *Analytical Chemistry* 92 (19) (2020) 13369–13377, <https://doi.org/10.1021/acs.analchem.0c02744>.
- [90] S.M. Copp, S.M. Swasey, A. Gorovits, P. Bogdanov, E.G. Gwinn, General Approach for Machine Learning-Aided Design of DNA-Stabilized Silver Clusters, *Chemistry of Materials* 32 (1) (2020) 430–437, <https://doi.org/10.1021/acs.chemmater.9b04040>.
- [91] S.T. Hill, R. Kuintzle, A. Teegarden, E. Merrill III, P. Danaee, D.A. Hendrix, A deep recurrent neural network discovers complex biological rules to decipher RNA protein-coding potential, *Nucleic Acids Research* 46 (16) (2018) 8105–8113, <https://doi.org/10.1093/nar/gky567> %J Nucleic Acids Research.
- [92] F. Zhai, Y. Guan, Y. Li, S. Chen, R. He, Predicting the Fluorescence Properties of Hairpin-DNA-Templated Silver Nanoclusters via Deep Learning, *ACS Applied Nano Materials* 5 (7) (2022) 9615–9624, <https://doi.org/10.1021/acsanm.2c01827>.
- [93] B. Wang, S. Lu, The light of carbon dots: From mechanism to applications, *Matter* 5 (1) (2022) 110–149, <https://doi.org/10.1016/j.matt.2021.10.016>.
- [94] L. Dordević, F. Arcudi, M. Cacioppo, M. Prato, A multifunctional chemical toolbox to engineer carbon dots for biomedical and energy applications, *Nature Nanotechnology* 17 (2) (2022) 112–130, <https://doi.org/10.1038/s41565-021-01051-7>.
- [95] B. Wang, J. Yu, L. Sui, S. Zhu, Z. Tang, B. Yang, S. Lu, Rational Design of Multi-Color-Emissive Carbon Dots in a Single Reaction System by Hydrothermal, *Advanced Science* 8 (1) (2021) 2001453, <https://doi.org/10.1002/adv.202001453>.
- [96] X. Yang, L. Ai, J. Yu, G.I.N. Waterhouse, L. Sui, J. Ding, B. Zhang, X. Yong, S. Lu, Photoluminescence mechanisms of red-emissive carbon dots derived from non-conjugated molecules, *Science Bulletin* 67 (14) (2022) 1450–1457, <https://doi.org/10.1016/j.scib.2022.06.013>.
- [97] Y. Luo, S. Bag, O. Zaremba, A. Cierpka, J. Andreo, S. Wuttke, P. Friederich, M. Tsotsalas, MOF Synthesis Prediction Enabled by Automatic Data Mining and Machine Learning**, *Angewandte Chemie International Edition* 61 (19) (2022), <https://doi.org/10.1002/anie.202200242>.
- [98] M.Y. Lu, T.Y. Chen, D.F.K. Williamson, M. Zhao, M. Shady, J. Lipkova, F. Mahmood, AI-based pathology predicts origins for cancers of unknown primary, *Nature* 594 (7861) (2021) 106–110, <https://doi.org/10.1038/s41586-021-03512-4>.
- [99] X.-Y. Wang, B.-B. Chen, J. Zhang, Z.-R. Zhou, J. Lv, X.-P. Geng, R.-C. Qian, Exploiting deep learning for predictable carbon dot design, *Chemical Communications* 57 (4) (2021) 532–535, <https://doi.org/10.1039/D0CC07882D>.
- [100] J. Chen, J.B. Luo, M.Y. Hu, J. Zhou, C.Z. Huang, H. Liu, Controlled Synthesis of Multicolor Carbon Dots Assisted by Machine Learning, *Advanced Functional Materials* 33 (2) (2023) 2210095, <https://doi.org/10.1002/adfm.202210095>.
- [101] J.-H. Park, A.-Y. Lu, M.M. Tavakoli, N.Y. Kim, M.-H. Chiu, H. Liu, T. Zhang, Z. Wang, J. Wang, L.G.P. Martins, Z. Luo, M. Chi, J. Miao, J. Kong, Revealing Variable Dependencies in Hexagonal Boron Nitride Synthesis via Machine Learning, *Nano Letters* 23 (11) (2023) 4741–4748, <https://doi.org/10.1021/acs.nanolett.2c04624>.
- [102] T.V. Galassi, P.V. Jena, J. Shah, G. Ao, E. Molitor, Y. Bram, A. Frankel, J. Park, J. Jessurun, D.S. Ory, A. Haimovitz-Friedman, D. Roxbury, J. Mittal, M. Zheng, R. E. Schwartz, D.A. Heller, An optical nanoreporter of endolysosomal lipid accumulation reveals enduring effects of diet on hepatic macrophages in vivo, *Science Translational Medicine* 10 (461) (2018) eaar2680.
- [103] S. Dasgupta, T. Auth, G. Gompfer, Shape and Orientation Matter for the Cellular Uptake of Nonpherical Particles, *Nano Letters* 14 (2) (2014) 687–693, <https://doi.org/10.1021/nl403949h>.
- [104] M. Gravely, M.M. Safaei, D. Roxbury, Biomolecular Functionalization of a Nanomaterial To Control Stability and Retention within Live Cells, *Nano Letters* 19 (9) (2019) 6203–6212, <https://doi.org/10.1021/acs.nanolett.9b02267>.
- [105] S. Jin, P. Wijesekara, P.D. Boyer, K.N. Dahl, M.F. Islam, Length-dependent intracellular bundling of single-walled carbon nanotubes influences retention, *Journal of Materials Chemistry B* 5 (32) (2017) 6657–6665, <https://doi.org/10.1039/C7TB00735C>.
- [106] Z. Gao, J.A. Varela, L. Groc, B. Lounis, L. Cognet, Toward the suppression of cellular toxicity from single-walled carbon nanotubes, *Biomaterials Science* 4 (2) (2016) 230–244, <https://doi.org/10.1039/C5BM00134J>.
- [107] M. Gravely, D. Roxbury, Multispectral Fingerprinting Resolves Dynamics of Nanomaterial Trafficking in Primary Endothelial Cells, *ACS Nano* 15 (7) (2021) 12388–12404, <https://doi.org/10.1021/acs.nano.1c04500>.
- [108] A. Ismail, H.-L. Truong, W. Kastner, Manufacturing process data analysis pipelines: a requirements analysis and survey, *Journal of Big Data* 6 (1) (2019) 1, <https://doi.org/10.1186/s40537-018-0162-3>.
- [109] T. Zhang, P. Li, H. Yu, F. Wang, X. Wang, T. Yang, W. Yang, W.J. Li, Y. Wang, L. Liu, Fabrication of flexible microlens arrays for parallel super-resolution imaging, *Applied Surface Science* 504 (2020) 144375, <https://doi.org/10.1016/j.apsusc.2019.144375>.
- [110] P. Liu, Z. Tian, K. Yang, T.D. Naquin, N. Hao, H. Huang, J. Chen, Q. Ma, H. Bachman, P. Zhang, X. Xu, J. Hu, T.J. Huang, Acoustofluidic black holes for multifunctional in-droplet particle manipulation, *Science Advances* 8 (13) (2022) eaab2592.
- [111] G. Jin, S. Hong, J. Rich, J. Xia, K. Kim, L. You, C. Zhao, T.J. Huang, Intelligent nanoscope for rapid nanomaterial identification and classification, *Lab on a Chip* 22 (16) (2022) 2978–2985, <https://doi.org/10.1039/D2LC00026J>.
- [112] M. Levy, K. Fujihara, J. Palace, New therapies for neuromyelitis optica spectrum disorder, *The Lancet Neurology* 20 (1) (2021) 60–67, [https://doi.org/10.1016/S1474-4422\(20\)30392-6](https://doi.org/10.1016/S1474-4422(20)30392-6).
- [113] W. Huang, J. ZhangBao, X. Chang, L. Wang, C. Zhao, J. Lu, M. Wang, X. Ding, Y. Xu, L. Zhou, D. Li, M.K. Behne, J.M. Behne, M.R. Yeaman, E. Katz, C. Lu, C. Quan, Neuromyelitis optica spectrum disorder in China: Quality of life and medical care experience, *Multiple Sclerosis and Related Disorders* 46 (2020) 102542, <https://doi.org/10.1016/j.msard.2020.102542>.
- [114] S.J. Pittcock, A. Zekeridou, B.G. Weinschenker, Hope for patients with neuromyelitis optica spectrum disorders — from mechanisms to trials, *Nature Reviews Neurology* 17 (12) (2021) 759–773, <https://doi.org/10.1038/s41582-021-00568-8>.
- [115] W. Chen, H. Yu, Y. Hao, W. Liu, R. Wang, Y. Huang, J. Wu, L. Feng, Y. Guan, L. Huang, K. Qian, Comprehensive Metabolic Fingerprints Characterize Neuromyelitis Optica Spectrum Disorder by Nanoparticle-Enhanced Laser Desorption/Ionization Mass Spectrometry, *ACS Nano* 17 (20) (2023) 19779–19792, <https://doi.org/10.1021/acsnano.3c03765>.
- [116] B. Demirbay, D. Bayram Kara, Ş. Uğur, Multivariate regression (MVR) and different artificial neural network (ANN) models developed for optical transparency of conductive polymer nanocomposite films, *Expert Systems with Applications* (2022) 117937.
- [117] A. Barkhordari, H.R. Mashayekhi, P. Amiri, S. Özçelik, F. Hanife, Y. Azizian-Kalanderagh, Performance of machine learning algorithms in spectroscopic ellipsometry data analysis of ZnTiO₃ nanocomposite, *Scientific Reports* 14 (1) (2024) 1617, <https://doi.org/10.1038/s41598-023-50620-4>.
- [118] A. Rafiei Miandashli, L. Khosravi Khorashad, M.E. Kordesch, A.O. Govorov, H.H. Richardson, Experimental and Theoretical Observation of Photothermal Chirality in Gold Nanoparticle Helicoids, *ACS Nano* 14(4) (2020) 4188–4195, <https://doi.org/10.1021/acsnano.9b09062>.
- [119] H. Wang, L. Yang, A. Tian, B. Huang, Fractal recognition and contact characteristics of ionic electroactive polymer interface based on microstructure analysis, *Tribology International* 186 (2023) 108650, <https://doi.org/10.1016/j.triboint.2023.108650>.
- [120] J.M. Ede, Deep learning in electron microscopy, *Machine Learning: Science and Technology* 2 (1) (2021) 011004, <https://doi.org/10.1088/2632-2153/abd614>.
- [121] Z. Jin, P. Li, Y. Meng, Z. Fang, D. Xiao, G. Yu, Understanding the inter-site distance effect in single-atom catalysts for oxygen electroreduction, *Nature Catalysis* 4 (7) (2021) 615–622, <https://doi.org/10.1038/s41929-021-00650-w>.
- [122] Z.-Y. Wu, M. Karamad, X. Yong, Q. Huang, D.A. Cullen, P. Zhu, C. Xia, Q. Xiao, M. Shakouri, F.-Y. Chen, J.Y. Kim, Y. Xia, K. Heck, Y. Hu, M.S. Wong, Q. Li, I. Gates, S. Siahrostami, H. Wang, Electrochemical ammonia synthesis via nitrate reduction on Fe single atom catalyst, *Nature Communications* 12 (1) (2021) 2870, <https://doi.org/10.1038/s41467-021-23115-x>.
- [123] H.T. Chung, D.A. Cullen, D. Higgins, B.T. Sneed, E.F. Holby, K.L. More, P. Zelenay, Direct atomic-level insight into the active sites of a high-performance PGM-free ORR catalyst, *Science* 357 (6350) (2017) 479–484.
- [124] H. Jia, C. Wang, C. Wang, P. Clancy, Machine Learning Approach to Enable Spectral Imaging Analysis for Particularly Complex Nanomaterial Systems, *ACS Nano* 17 (1) (2023) 453–460, <https://doi.org/10.1021/acsnano.2c08884>.
- [125] N. Oinonen, C. Xu, B. Alldritt, F.F. Canova, F. Urtev, S. Cai, O. Krejčí, J. Kannala, P. Liljeroth, A.S. Foster, Electrostatic Discovery Atomic Force Microscopy, *ACS Nano* 16 (1) (2022) 89–97, <https://doi.org/10.1021/acsnano.1c06840>.
- [126] Y.F. Dufrène, T. Ando, R. Garcia, D. Alsteens, D. Martinez-Martin, A. Engel, C. Gerber, D.J. Müller, Imaging modes of atomic force microscopy for application in molecular and cell biology, *Nature Nanotechnology* 12 (4) (2017) 295–307, <https://doi.org/10.1038/nnano.2017.45>.
- [127] B. Alldritt, P. Hapala, N. Oinonen, F. Urtev, O. Krejčí, F. Federici Canova, J. Kannala, F. Schulz, P. Liljeroth, A.S. Foster, Automated structure discovery in atomic force microscopy, *Science Advances* 6 (9) (2020) eaay6913.
- [128] Y.-J. Kim, J. Lim, D.-N. Kim, Accelerating AFM Characterization via Deep-Learning-Based Image Super-Resolution, *Small* 18 (3) (2022) 2103779, <https://doi.org/10.1002/smll.202103779>.
- [129] S. Zhu, Q. Zhou, M. Wang, J. Dale, Z. Qiang, Y. Fan, M. Zhu, C. Ye, Modulating electromagnetic interference shielding performance of ultra-lightweight composite foams through shape memory function, *Composites Part B: Engineering* 204 (2021) 108497, <https://doi.org/10.1016/j.compositesb.2020.108497>.
- [130] B. Zhao, J. Deng, C. Zhao, C. Wang, Y.G. Chen, M. Hamidinejad, R. Li, C.B. Park, Achieving wideband microwave absorption properties in PVDF nanocomposite foams with an ultra-low MWCNT content by introducing a microcellular structure, *Journal of Materials Chemistry C* 8 (1) (2020) 58–70, <https://doi.org/10.1039/C9TC04575A>.
- [131] C.-H. Cui, D.-X. Yan, H. Pang, L.-C. Jia, X. Xu, S. Yang, J.-Z. Xu, Z.-M. Li, A high heat-resistance bioplastic foam with efficient electromagnetic interference shielding, *Chemical Engineering Journal* 323 (2017) 29–36, <https://doi.org/10.1016/j.cej.2017.04.050>.
- [132] W. Samek, A. Binder, G. Montavon, S. Lapuschkin, K.R. Müller, Evaluating the Visualization of What a Deep Neural Network Has Learned, *IEEE Transactions on Neural Networks and Learning Systems* 28 (11) (2017) 2660–2673, <https://doi.org/10.1109/TNNLS.2016.2599820>.
- [133] M. Shi, C.-P. Feng, Y.-L. Tu, G.-S. Shi, P.-Y. He, Y. Zhang, J. Zhang, J. Li, S. Guo, Visualization of Deep Convolutional Neural Networks to Investigate Porous Nanocomposites for Electromagnetic Interference Shielding, *ACS Applied Materials & Interfaces* 15 (18) (2023) 22602–22615, <https://doi.org/10.1021/acsami.3c04557>.
- [134] M.C. di Gregorio, L.J.W. Shimon, V. Brumfeld, L. Houben, M. Lahav, M.E. van der Boom, Emergence of chirality and structural complexity in single crystals at the molecular and morphological levels, *Nature Communications* 11 (1) (2020) 380, <https://doi.org/10.1038/s41467-019-13925-5>.

- [135] A. Ben-Moshe, A. da Silva, A. Müller, A. Abu-Odeh, P. Harrison, J. Waelder, F. Niroui, C. Ophus, A.M. Minor, M. Asta, W. Theis, P. Ercius, A.P. Alivisatos, The chain of chirality transfer in tellurium nanocrystals, *Science* 372 (6543) (2021) 729–733.
- [136] C.K. Groschner, A.J. Pattison, A. Ben-Moshe, A.P. Alivisatos, W. Theis, M.C. Scott, Classifying handedness in chiral nanomaterials using label error robust deep learning, *Npj Computational Materials* 8 (1) (2022) 149, <https://doi.org/10.1038/s41524-022-00822-7>.
- [137] C. Cabrera, D. Cervantes, F. Muñoz, G. Hirata, P. Juárez, D.-L. Flores, Classifying nanostructured and heterogeneous materials from transmission electron microscopy images using convolutional neural networks, *Neural Computing and Applications* 34 (13) (2022) 11035–11047, <https://doi.org/10.1007/s00521-022-07029-3>.
- [138] L. Yang, H. Wang, D. Zhang, Y. Yang, D. Leng, Large deformation, high energy density dielectric elastomer actuators: Principles, factors, optimization, applications, and prospects, *Chemical Engineering Journal* 489 (2024) 151402, <https://doi.org/10.1016/j.cej.2024.151402>.
- [139] L. Yang, H. Wang, Y. Yang, Y. Li, Self-healing cellulose-based hydrogels: From molecular design to multifarious applications, *Carbohydrate Polymers* 347 (2025) 122738.
- [140] K. Baek, H. Park, H. Shin, S. Yang, M. Cho, Multiscale modeling to evaluate combined effect of covalent grafting and clustering of silica nanoparticles on mechanical behaviors of polyimide matrix composites, *Composites Science and Technology* 206 (2021) 108673, <https://doi.org/10.1016/j.compscitech.2021.108673>.
- [141] K. Baek, T. Hwang, W. Lee, H. Chung, M. Cho, Deep learning aided evaluation for electromechanical properties of complexly structured polymer nanocomposites, *Composites Science and Technology* 228 (2022) 109661, <https://doi.org/10.1016/j.compscitech.2022.109661>.
- [142] J. Kim, S. Saremi, M. Acharya, G. Velarde, E. Parsonnet, P. Donahue, A. Qualls, D. Garcia, L.W. Martin, Ultrahigh capacitive energy density in ion-bombarded relaxor ferroelectric films, *Science* 369 (6499) (2020) 81–84.
- [143] H. Wang, Y. Yang, L. Yang, Optimization of dielectric loss in calcium copper titanate based on different doping modification strategies, *Ceramics International* 49(23, Part B) (2023) 38399–38419, <https://doi.org/https://doi.org/10.1016/j.ceramint.2023.09.186>.
- [144] T. Zhang, X. Chen, Y. Thakur, B. Lu, Q. Zhang, J. Runt, Q.M. Zhang, A highly scalable dielectric metamaterial with superior capacitor performance over a broad temperature, *Science Advances* 6 (4) (2020) eaax6622.
- [145] L. Yang, H. Wang, S. Fang, M. Li, Research progress on energy storage performance enhancement strategies for polyvinylidene fluoride-based composites, *Journal of Alloys and Compounds* 960 (2023) 170831, <https://doi.org/10.1016/j.jallcom.2023.170831>.
- [146] J. Liu, L. Ji, J. Yu, S. Ding, S. Luo, B. Chu, J. Xu, R. Sun, S. Yu, Enhanced breakdown strength and electrostatic energy density of polymer nanocomposite films realized by heterostructure ZnO-ZnS nanoparticles, *Chemical Engineering Journal* 456 (2023) 140950, <https://doi.org/10.1016/j.cej.2022.140950>.
- [147] Y. Jiang, X. Zhang, Z. Shen, X. Li, J. Yan, B.-W. Li, C.-W. Nan, Ultrahigh Breakdown Strength and Improved Energy Density of Polymer Nanocomposites with Gradient Distribution of Ceramic Nanoparticles, *Advanced Functional Materials* 30 (4) (2020) 1906112, <https://doi.org/10.1002/adfm.201906112>.
- [148] H. Li, D. Ai, L. Ren, B. Yao, Z. Han, Z. Shen, J. Wang, L.-Q. Chen, Q. Wang, Scalable Polymer Nanocomposites with Record High-Temperature Capacitive Performance Enabled by Rationally Designed Nanostructured Inorganic Fillers, *Advanced Materials* 31 (23) (2019) 1900875, <https://doi.org/10.1002/adma.201900875>.
- [149] Z.-H. Shen, Z.-W. Bao, X.-X. Cheng, B.-W. Li, H.-X. Liu, Y. Shen, L.-Q. Chen, X.-G. Li, C.-W. Nan, Designing polymer nanocomposites with high energy density using machine learning, *Npj Computational Materials* 7 (1) (2021) 110, <https://doi.org/10.1038/s41524-021-00578-6>.
- [150] J. Liu, Y. Zhang, Y. Zhang, S. Kitipornchai, J. Yang, Machine learning assisted prediction of mechanical properties of graphene/aluminium nanocomposite based on molecular dynamics simulation, *Materials & Design* 213 (2022) 110334, <https://doi.org/10.1016/j.matdes.2021.110334>.
- [151] B. Bahtiri, B. Arash, S. Scheffler, M. Jux, R. Rolfes, A machine learning-based viscoelastic-viscoplastic model for epoxy nanocomposites with moisture content, *Computer Methods in Applied Mechanics and Engineering* 415 (2023) 116293, <https://doi.org/10.1016/j.cma.2023.116293>.
- [152] S. Deka, F. Mozafari, A. Mallick, Microstructural, mechanical, tribological, and corrosion behavior of ultrafine bio-degradable Mg/CeO₂ nanocomposites: Machine learning-based modeling and experiment, *Tribology International* 190 (2023) 109063, <https://doi.org/10.1016/j.triboint.2023.109063>.
- [153] S.P. Surwade, S.N. Smirnov, I.V. Vlassiok, R.R. Unocic, G.M. Veith, S. Dai, S. M. Mahurin, Water desalination using nanoporous single-layer graphene, *Nature Nanotechnology* 10 (5) (2015) 459–464, <https://doi.org/10.1038/nnano.2015.37>.
- [154] M. Heiranian, A.B. Farimani, N.R. Aluru, Water desalination with a single-layer MoS₂ nanopore, *Nature Communications* 6 (1) (2015) 8616, <https://doi.org/10.1038/ncomms9616>.
- [155] Z. Yao, B. Sánchez-Lengeling, N.S. Bobbitt, B.J. Bucior, S.G.H. Kumar, S.P. Collins, T. Burns, T.K. Woo, O.K. Farha, R.Q. Snurr, A. Aspuru-Guzik, Inverse design of nanoporous crystalline reticular materials with deep generative models, *Nature Machine Intelligence* 3 (1) (2021) 76–86, <https://doi.org/10.1038/s42256-020-00271-1>.
- [156] M. Karamad, R. Magar, Y. Shi, S. Siahrostami, I.D. Gates, A. Barati Farimani, Orbital graph convolutional neural network for material property prediction, *Physical Review Materials* 4(9) (2020) 093801, <https://doi.org/10.1103/PhysRevMaterials.4.093801>.
- [157] A. Krizhevsky, I. Sutskever, G.E. Hinton, ImageNet classification with deep convolutional neural networks, *Communications of the ACM* 60(6 %J Commun, ACM) (2017) 84–90, <https://doi.org/10.1145/3065386>.
- [158] Y. Wang, Z. Cao, A. Barati Farimani, Efficient water desalination with graphene nanopores obtained using artificial intelligence, *Npj 2D Materials and Applications* 5 (1) (2021) 66, <https://doi.org/10.1038/s41699-021-00246-9>.
- [159] W. Ji, S. Deng, Autonomous Discovery of Unknown Reaction Pathways from Data by Chemical Reaction Neural Network, *The Journal of Physical Chemistry A* 125 (4) (2021) 1082–1092, <https://doi.org/10.1021/acs.jpca.0c09316>.
- [160] J.S. Owen, E.M. Chan, H. Liu, A.P. Alivisatos, Precursor Conversion Kinetics and the Nucleation of Cadmium Selenide Nanocrystals, *Journal of the American Chemical Society* 132 (51) (2010) 18206–18213, <https://doi.org/10.1021/ja106777j>.
- [161] J.C. Dahl, X. Wang, X. Huang, E.M. Chan, A.P. Alivisatos, Elucidating the Weakly Reversible Cs–Pb–Br Perovskite Nanocrystal Reaction Network with High-Throughput Maps and Transformations, *Journal of the American Chemical Society* 142 (27) (2020) 11915–11926, <https://doi.org/10.1021/jacs.0c04997>.
- [162] C. Rackauckas, Q. Nie, DifferentialEquations.jl – A Performant and Feature-Rich Ecosystem for Solving Differential Equations in Julia, *Journal of Open Research Software* (2017), <https://doi.org/10.5334/jors.151>.
- [163] J.C. Dahl, S. Niblett, Y. Cho, X. Wang, Y. Zhang, E.M. Chan, A.P. Alivisatos, Scientific Machine Learning of 2D Perovskite Nanosheet Formation, *Journal of the American Chemical Society* 145 (42) (2023) 23076–23087, <https://doi.org/10.1021/jacs.3c05984>.
- [164] C. Wang, L. Wang, A. Soo, N. Bansidhar Pathak, H. Kyong Shon, Machine learning based prediction and optimization of thin film nanocomposite membranes for organic solvent nanofiltration, *Separation and Purification Technology* 304 (2023) 122328, <https://doi.org/10.1016/j.seppur.2022.122328>.
- [165] W. Hu, L. Wan, Y. Jian, C. Ren, K. Jin, X. Su, X. Bai, H. Haick, M. Yao, W. Wu, Electronic Noses: From Advanced Materials to Sensors Aided with Data Processing, *Advanced Materials Technologies* 4 (2) (2019) 1800488, <https://doi.org/10.1002/admt.201800488>.
- [166] L. Cai, X. Dong, G. Wu, J. Sun, N. Chen, H. Wei, S. Zhu, Q. Tian, X. Wang, Q. Jing, P. Li, B. Liu, Ultrasensitive acetone gas sensor can distinguish the diabetic state of people and its high performance analysis by first-principles calculation, *Sensors and Actuators b: Chemical* 351 (2022) 130863, <https://doi.org/10.1016/j.snb.2021.130863>.
- [167] J.F.d.S. Petruc, D.N. Barreto, M.A. Dias, E.P. Felix, A.A. Cardoso, Analytical methods applied for ozone gas detection: A review, *TrAC Trends in Analytical Chemistry* 149 (2022) 116552, <https://doi.org/https://doi.org/10.1016/j.trac.2022.116552>.
- [168] H. Chai, Z. Zheng, K. Liu, J. Xu, K. Wu, Y. Luo, H. Liao, M. Debligny, C. Zhang, Stability of Metal Oxide Semiconductor Gas Sensors: A Review, *IEEE Sensors Journal* 22 (6) (2022) 5470–5481, <https://doi.org/10.1109/JSEN.2022.3148264>.
- [169] Y. Cheng, L. Li, Z. Liu, S. Yan, F. Cheng, Y. Yue, S. Jia, J. Wang, Y. Gao, L. Li, 3D Porous MXene Aerogel through Gas Foaming for Multifunctional Pressure Sensor, *Research* 2022 (2022) 9843268.
- [170] X. Liu, W. Zheng, R. Kumar, M. Kumar, J. Zhang, Conducting polymer-based nanostructures for gas sensors, *Coordination Chemistry Reviews* 462 (2022) 214517, <https://doi.org/10.1016/j.ccr.2022.214517>.
- [171] D. Wang, D. Zhang, M. Tang, H. Zhang, T. Sun, C. Yang, R. Mao, K. Li, J. Wang, Ethylene chlorotrifluoroethylene/hydrogel-based liquid-solid triboelectric nanogenerator driven self-powered MXene-based sensor system for marine environmental monitoring, *Nano Energy* 100 (2022) 107509, <https://doi.org/10.1016/j.nanoen.2022.107509>.
- [172] C. Ma, M.-G. Ma, C. Si, X.-X. Ji, P. Wan, Flexible MXene-Based Composites for Wearable Devices, *Advanced Functional Materials* 31 (22) (2021) 2009524, <https://doi.org/10.1002/adfm.202009524>.
- [173] R. Zhu, Y. Zhang, H. Zhong, X. Wang, H. Xiao, Y. Chen, X. Li, High-performance room-temperature NO₂ sensors based on CH₃NH₃PbBr₃ semiconducting films: Effect of surface capping by alkyl chain on sensor performance, *Journal of Physics and Chemistry of Solids* 129 (2019) 270–276, <https://doi.org/10.1016/j.jpcs.2019.01.020>.
- [174] J. Hu, D. Liu, X. Xia, B. Wang, D. Pan, Y. Cheng, Y. Lu, MXene/perovskite-based bionic human odor sensor array with machine learning, *Chemical Engineering Journal* 468 (2023) 143752, <https://doi.org/10.1016/j.cej.2023.143752>.
- [175] S. Sudarshan, S. Harikrishnan, G. RathiBhuvaneswari, V. Alamelu, S. Aanand, A. Rajasekar, M. Govarthanan, Impact of textile dyes on human health and bioremediation of textile industry effluent using microorganisms: current status and future prospects, *Journal of Applied Microbiology* 134 (2) (2022), <https://doi.org/10.1093/jambio/ixac064>.
- [176] M. Suresh, A. Sivasamy, Fabrication of graphene nanosheets decorated by nitrogen-doped ZnO nanoparticles with enhanced visible photocatalytic activity for the degradation of Methylene Blue dye, *Journal of Molecular Liquids* 317 (2020) 114112, <https://doi.org/10.1016/j.molliq.2020.114112>.
- [177] S.A. Ganiyu, M.A. Suleiman, W.A. Al-Amrani, A.K. Usman, S.A. Onaizi, Adsorptive removal of organic pollutants from contaminated waters using zeolitic imidazolate framework Composites: A comprehensive and Up-to-date review, *Separation and Purification Technology* 318 (2023) 123765, <https://doi.org/10.1016/j.seppur.2023.123765>.
- [178] S. Zhou, L. Xia, Z. Fu, C. Zhang, X. Duan, S. Zhang, Y. Wang, C. Ding, X. Liu, W. Xu, Purification of dye-contaminated ethanol-water mixture using magnetic cellulose powders derived from agricultural waste biomass, *Carbohydrate Polymers* 258 (2021) 117690, <https://doi.org/10.1016/j.carbpol.2021.117690>.

- [179] G. Alam, I. Ihsanullah, M. Naushad, M. Sillanpää, Applications of artificial intelligence in water treatment for optimization and automation of adsorption processes: Recent advances and prospects, *Chemical Engineering Journal* 427 (2022) 130011. <https://doi.org/10.1016/j.cej.2021.130011>.
- [180] J.A. Okolie, S. Savage, C.C. Ogbaga, B. Gunes, Assessing the potential of machine learning methods to study the removal of pharmaceuticals from wastewater using biochar or activated carbon, *Total Environment Research Themes* 1–2 (2022) 100001. <https://doi.org/10.1016/j.totert.2022.100001>.
- [181] G.M. BinMakhashen, S.A. Bahadi, H.A. Al-Jamimi, S.A. Onaizi, Ensemble meta machine learning for predicting the adsorption of anionic and cationic dyes from aqueous solutions using Polymer/graphene/clay/MgFeAl-LTH nanocomposite, *Chemosphere* 349 (2024) 140861. <https://doi.org/10.1016/j.chemosphere.2023.140861>.
- [182] G. Ramkumar, M. Tamilselvi, S.D. Sundarsingh Jebaseelan, V. Mohanavel, H. Kamyab, G. Anitha, R. Thandaiah Prabu, M. Rajasimman, Enhanced machine learning for nanomaterial identification of photo thermal hydrogen production, *International Journal of Hydrogen Energy* 52 (2024) 696–708. <https://doi.org/10.1016/j.ijhydene.2023.07.128>.
- [183] E. Westphal, H. Seitz, A machine learning method for defect detection and visualization in selective laser sintering based on convolutional neural networks, *Additive Manufacturing* 41 (2021) 101965. <https://doi.org/10.1016/j.addma.2021.101965>.
- [184] H. Tao, T. Wu, M. Aldeghi, T.C. Wu, A. Aspuru-Guzik, E. Kumacheva, Nanoparticle synthesis assisted by machine learning, *Nature Reviews Materials* 6 (8) (2021) 701–716. <https://doi.org/10.1038/s41578-021-00337-5>.
- [185] F. Mekki-Berrada, Z. Ren, T. Huang, W.K. Wong, F. Zheng, J. Xie, I.P.S. Tian, S. Jayavelu, Z. Mahfoud, D. Bash, K. Hippalgaonkar, S. Khan, T. Buonassisi, Q. Li, X. Wang, Two-step machine learning enables optimized nanoparticle synthesis, *Npj Computational Materials* 7 (1) (2021) 55. <https://doi.org/10.1038/s41524-021-00520-w>.